MUON ON THE QUOTIENT: GAUGE-EQUIVARIANT DESCENT FOR HOMOGENEOUS NETWORKS

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ABSTRACT

Positively homogeneous networks, including ReLU networks and the SwiGLU blocks of modern transformers, have a rescaling symmetry. That is, scaling up the incoming weights of a hidden unit and scaling down its outgoing weights leaves the function unchanged. Orthogonalized optimizers such as Muon do not respect this symmetry, thereby equivalent parameterizations of the same function receive different updates, and a function-preserving rescaling can lead Muon to a sharper basin with a higher loss. We introduce Gauge-Muon, a family of optimizers that keeps Muon's orthogonalized update but applies it to gradients rescaled by per-unit factors computed from the weights, and then undoes the rescaling. Every member of the family is exactly equivariant under hidden-unit rescaling at every finite step size, and performs exact steepest descent in the quotient geometry that identifies all equivalent parameterizations. We further show how the family's exponent allocates functional motion across hidden units, and derive an invariant capacity bound on generalization. Empirically, Gauge-Muon is unaffected by rescalings that degrade Muon and improves on Muon in image classification and GPT-2 language modeling, with the most consistent gains in mixture-of-experts models with narrow experts.

1 INTRODUCTION

Orthogonalized optimizers such as Muon (Jordan et al., 2024) have become an important alternative to Adam (Kingma & Ba, 2014) for training deep networks. Iterative approximations such as Newton–Schulz iterations (Björck & Bowie, 1971; Amsel et al., 2026) make Muon cheap enough to use in LLM pretraining (Liu et al., 2025; Bai et al., 2026). The core idea of Muon is to perform steepest descent under the spectral norm (Bernstein & Newhouse, 2024b). More generally, one can design an optimizer by choosing a norm on parameter space and solving the corresponding linear minimization oracle (Large et al., 2024; Bernstein & Newhouse, 2024a; Pethick et al., 2025). This view carries an implicit assumption that the weight matrix is the right object on which to place a norm.

ReLU networks and the SwiGLU feed-forward blocks of modern transformers (Vaswani et al., 2023; Shazeer, 2020) have a rescaling symmetry. That is, scaling up the incoming weights of a hidden unit and scaling down its outgoing weights by the same factor leaves the function unchanged. In a layer, this corresponds to multiplying the two weight matrices by a positive diagonal matrix and its inverse. The polar map of Muon does not commute with such rescalings, thereby Muon assigns different updates to different parameterizations of the *same* function. These rescalings can occur in practice from initialization and from the per-channel scale folding used in quantization (Nagel et al., 2019; Xiao et al., 2022). After a function-preserving rescaling at initialization, Muon converges to a sharper basin with a higher loss (Figure 1, Appendix G.1).

In this paper, we ask: *can we design an optimizer that keeps Muon's spectral oracle, the polar map, while guaranteeing that its updates depend only on the function a network represents and not on how that function is parameterized?* To answer this, we introduce preconditioners that transform together with the rescaling. They are built from the norms of each hidden unit's incoming and outgoing weights and from its path mass which is the product of these two norms. This path mass is invariant under any rescaling. Applying the preconditioners to the gradients yields covectors that are identical across all parameterizations of the same function. To this end, we introduce *Gauge-Muon* by applying the polar map to these covectors and maps the result back through the same preconditioners

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to form the update. Existing rescaling-invariant methods obtain invariance in other ways such as through a path-normalized Euclidean geometry (Neyshabur et al., 2015a), by optimizing directly in invariant path coordinates (Meng et al., 2019), by moving to a well-conditioned representative of each set of equivalent parameters (Stock et al., 2019; Lebeurrier et al., 2026), or by descending in canonically scaled coordinates (Uhlmann, 2026). None of them keeps the matrix-level coupling of an orthogonalized update. Conversely, symmetry-aware analyses of Muon (Singh, 2026) concern orthogonal changes of basis, which the polar map already respects, rather than rescaling. Closest to our work, Shirodkar (2026) make an adaptive optimizer equivariant by splitting its state into a component along the symmetry orbit and a component across it, measured in an invariant metric, and projecting the update back across the orbit. For the rescaling symmetry, however, that construction runs on an Adam base. It argues that orthogonalization removes the per-channel scale, so its Muon variant is made equivariant only to attention-head rotations. Gauge-Muon takes a different route. It uses no projection and no second-moment split. Instead it rescales the gradient by per-unit factors computed from the weights before the polar map and undoes them afterwards, thereby Muon's spectral update is itself equivariant to hidden-unit rescaling. To our knowledge, Gauge-Muon is the first optimizer that combines exact rescaling invariance with matrix-level orthogonalized updates.

Our contributions are as follows. (*i*) We introduce Gauge-Muon, a family of optimizers that are exactly equivariant under hidden-unit rescaling at every finite step size (Theorem 4.2). (*ii*) We show that Gauge-Muon is exact steepest descent under a state-dependent operator norm that descends to the quotient of parameter space by the rescaling group (Theorem 5.1). (*iii*) We characterize how the family's exponent allocates functional mobility across paths of different strength (Proposition 6.1) and derive an invariant capacity bound on generalization (Theorem F.1). (*iv*) We validate Gauge-Muon on image classification and GPT-2 language modeling, where it improves on Muon in all image settings and in every language-modeling setting at the low token budget; at the higher budget the gain persists in mixture-of-experts models with narrow experts.

2 Related Work

Spectral and orthogonalized optimizers. Muon (Jordan et al., 2024) replaces the momentum matrix by (an approximation to) its polar factor, computed with Newton-Schulz iterations (Björck & Bowie, 1971; Higham, 2008), and has been scaled to LLM pretraining (Liu et al., 2025). It descends from spectral descent (Carlson et al., 2015), is closely related to Shampoo (Gupta et al., 2018), and is steepest descent under the spectral norm (Bernstein & Newhouse, 2024b); modular norms (Large et al., 2024; Bernstein & Newhouse, 2024a) and linear-minimization-oracle (LMO) methods (Pethick et al., 2025) extend this view to whole networks. Muon is equivariant under orthogonal changes of basis, which commute with the polar map (Singh, 2026). All of these geometries are fixed norms on *parameter* space, and the polar map does not commute with positive diagonal congruences, so none is invariant under the hidden-unit rescaling. Our Gauge-Muon keeps the spectral LMO but applies it to state-dependent, gauge-invariant covectors.

Preconditioned norms. A steepest-descent step under a left/right-preconditioned operator norm can be computed by transforming the gradient, applying the base-norm LMO, and mapping back. This identity is implicit in the duality maps of Bernstein & Newhouse (2024b;a) and Pethick et al. (2025), and Veprikov et al. (2025) develop it into a general framework in which invariances are inherited from the transformation law of preconditioner. In this paper, our main focus is to propose a preconditioner. We derive the covariant factors from the *paired* incoming/outgoing gauge and path masses, prove exact finite-step equivariance. We show that the induced norm descends to a quotient norm on function space, giving a family of equivariant spectral geometries.

Rescaling-invariant optimization. There are existing studies on positive homogeneity of ReLU networks. Path-SGD (Neyshabur et al., 2015a) performs steepest descent under a rescaling-invariant path regularizer, G-SGD (Meng et al., 2019) optimizes in a basis of positively scale-invariant path variables, and Equi-normalization (Stock et al., 2019) selects a canonical representative of each orbit. Most recently, the unit-consistent adjoint framework (Uhlmann, 2026) obtains gauge-equivariant steepest descent by descending in canonically scaled coordinates. Such methods achieve invariance with entry-wise or Euclidean geometry, or by fixing one representative. In constrast, to our knowledge, our Gauge-Muon is the first approach to connect between exact invariance and a matrix-level spectral

LMO. Specifically, Gauge-Muon treats the balancing exponent α as a free geometric parameter, which yields a path-mobility degree of freedom that canonical Euclidean descent does not have.

Conservation laws and implicit bias. Under Euclidean gradient flow, rescaling symmetry conserves (Du et al., 2018; Kunin et al., 2020; Marcotte et al., 2023), and reparameterization-induced implicit bias is well described by mirror descent (Gunasekar et al., 2018; Li et al., 2022), with potentials that interpolate between entropic and Euclidean geometry as a function of the initialization scale or shape (Woodworth et al., 2019; Azulay et al., 2021). The conserved quantity depends on the gauge, so the implicit bias of these flows depends on the initial representative. Under Gauge-Muon, path masses follow a mirror flow under a Box–Cox potential indexed by the *optimizer* parameter α, independent of the gauge of the initialization.

3 Background and Motivation

In this section, we review the necessary background on the Muon optimizer, and explain how gauge symmetry motivates our approach. Note that notations used in the paper are provided in Appendix A

3.1 Muon from first principles

Here, we start with the exact matrix opertion that Muon optimizer (Jordan et al., 2024) tries to approximate.

The support polar factor Given $H \in \mathbb{R}^{m\times n}$ with compact SVD $H = P\Sigma Q^\top$, the *support polar factor* of H is defined as $\operatorname{polar}(H) = PQ^\top$. Here, only singular directions with positive singular value are involved. We can write the support polar factor in terms of the duality norm

$$\operatorname{polar}(H) \in \arg\max_{\|Q\|_{\mathrm{op}}\le 1} \langle H, Q\rangle_F, \qquad \langle H, \operatorname{polar}(H)\rangle_F = \|H\|_*. \tag{1}$$

This means that we can understand that $\operatorname{polar}(H)$ is the direction of largest first-order decrease if one is allowed a unit operator-norm step.

Consider $L(W)$ be differentiable loss function of a matrix parameter W, with gradient $G = \nabla_W L(W)$. One can show that among all perturbations ΔW satisfying $\|\Delta W\|_{\mathrm{op}} \le 1$, the most negative first-order change[1] is attained by $\Delta W = -\operatorname{polar}(G)$, and equals to $-\|G\|_*$. This is the gradient step that Muon optimiers wish to update model parameters.

Momentum. Practical implementations of Muon employ momentum. Specifically, given a momentum coefficient $\mu \in (0, 1)$, the momentum matrix is computed as an exponential moving average of the gradients G_t via the online update $M_t = \mu M_{t-1} + (1-\mu)G_t$. The Muon parameter update is then applied to the momentum matrix as $W_{t+1} = W_t - \eta \operatorname{polar}(M_t)$..

Newton-Schulz approximation. To avoid the computational cost of an explicit SVD, practical Muon implementations approximate the polar factor using a small number of Newton-Schulz iterations. This procedure progressively maps the nonzero singular values toward one using only matrix multiplications (Jordan et al., 2024). Throughout this paper, we use the exact polar factor in our theoretical derivations. Nevertheless, it may be replaced in practice by any deterministic approximation to the polar map.

3.2 Motivation

ReLU Network Our central subject in two-layer neural networks with ReLU activation

$$f_{U,V}(x) = V\,\mathrm{ReLU}(Ux) = \sum_{i=1}^{h} v_i\,\mathrm{ReLU}(u_i^\top x), \tag{2}$$

where $x \in \mathbb{R}^d$ is the input, $U \in \mathbb{R}^{h\times d}$ is the incoming matrix, $V \in \mathbb{R}^{k\times h}$ is the outgoing matrix, $u_i^\top$ denotes row i of U, and v_i denotes column i of V. Thus hidden unit i contributes $v_i\,\mathrm{ReLU}(u_i^\top x)$

[1] First-order change according to Talor approximation is $\langle G, \Delta W\rangle_F$

to the output. We define its incoming magnitude r_i and outgoing magnitude s_i and path mass p_i associated with hidden unit i

$$r_i = \|u_i\|_2, \quad s_i = \|v_i\|_2, \quad p_i = r_i s_i. \tag{3}$$

Here, the path mass p_i is understood as the product of the two side magnitudes r_i and s_i with a hidden route from input to output.

Function-preserving hidden-unit rescaling Let $D = \mathrm{diag}(d_1, \ldots, d_h)$ with every $d_i > 0$. Since the positive homogeneity of ReLU activation, $\mathrm{ReLU}(cz) = c\,\mathrm{ReLU}(z), c > 0$, we can write

$$f_{DU,VD^{-1}}(x) = VD^{-1}\,\mathrm{ReLU}(DUx) = V\,\mathrm{ReLU}(Ux) = f_{U,V}(x). \tag{4}$$

Therefore all pairs $(U, V) \sim (DU, VD^{-1})$ represent the same function output. Let $\mathcal{G} = (\mathbb{R}_{>0})^h$ be the set of all such D forms the positive diagonal group. We call this a *gauge symmetry* that move along the orbit generated by $\mathcal{G}$ changes coordinates but not the represented function. Under this action, the incomming, outgoing magnitudes and path mass is updated as $r'_i = d_i r_i$, $s'_i = d_i^{-1} s_i$, $p'_i = r'_i s'_i = p_i$. Thererfore, the path mass p_i is an exact gauge invariant.

Why ordinary Muon breaks the symmetry Let $\mathcal{L}(U, V)$ be any differentiable loss function that depends on (U, V) only through the predictor. We have $\mathcal{L}(DU, VD^{-1}) = \mathcal{L}(U, V)$. Let $G_U = \nabla_U \mathcal{L}(U, V)$ and $G_V = \nabla_V \mathcal{L}(U, V)$. At the transformed point $(U', V') = (DU, VD^{-1})$, the inverse coordinate map is $U = D^{-1}U'$ and $V = V'D$. Applying the chain rule gives $G'_U = D^{-1}G_U$ and $G'_V = G_V D$. This means incoming and outgoing gradients therefore transform in the *opposite* way from the corresponding parameters.

Suppose vanilla Muon optimizer is applied independently to the two blocks U, V in ReLU neural network. Gauge equivariance may require its incoming update to transform as $\Delta U' = D\Delta U$. However, the raw gradient transforms as $D^{-1}G_U$, and in general, we have $\mathrm{polar}(D^{-1}G_U) \neq D\ \mathrm{polar}(G_U)$. This mismatch is exactly analogous to the classical observation that Euclidean SGD is not rescaling invariant in ReLU networks (Neyshabur et al., 2015a). Muon optimizer solves optimization setting involved matrices. However, in the setting of ReLU network, those matrices are gauge-dependent objects. For an illustration, Figure 1 shows how the loss landscapes (Li et al., 2018) of different optimizers change under gauge rescaling. The loss landscape of Muon and Adam optimizer change under the rescaling while our Gauge-Muon is invariant under gauge. A detailed discusion on the sharpness of their loss landscape (Dinh et al., 2017) is provided in Appendix G.1.

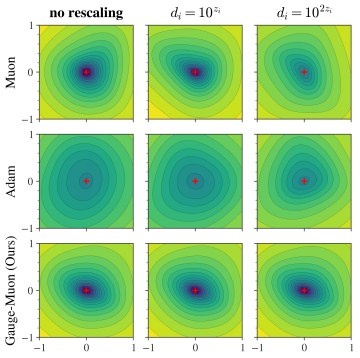


Figure 1: Loss landscape of three optimizers under diagonal rescaling $D = \mathrm{diag}(d_1, \ldots, d_h), d_i = 10^{z_i}_i$ or $d_i = 10^{2z_i}$ with $z_i \sim \mathcal{U}(-1, 1)$. See Appendix G.1 for a detailed discussion including the sharpness of loss landscape (Dinh et al., 2017).

4 Gauge-Muon Optimizer

Motivated by the gauge symmetry of ReLU networks, we now introduce Gauge-Muon. Rather than treating the incoming and outgoing weights of a hidden layer as independent matrices, Gauge-Muon updates them jointly as a pair (U, V). We first construct *covariant* scaling factors that combine information from both U and V.

Construct covariant scaling factors Let us define $A_\alpha = \mathrm{diag}\big(r_i p_i^{-\alpha}\big)$ and $B_\alpha = \mathrm{diag}\big(s_i p_i^{-\alpha}\big)$ for an exponent $\alpha \in \mathbb{R}$. Because p_i is invariant while r_i and s_i transform oppositely, $A'_\alpha = DA_\alpha$, $B'_\alpha = D^{-1}B_\alpha$. The transformed gradient blocks $H_U = A_\alpha G_U$, $H_V = G_V B_\alpha$ are therefore invariant:

$$H'_U = A'_\alpha G'_U = (DA_\alpha)(D^{-1}G_U) = H_U, \quad H'_V = G'_V B'_\alpha = (G_V D)(D^{-1}B_\alpha) = H_V. \tag{5}$$

We call H_U, H_V the *canonical covectors*. The "canonical" here means gauge independent.

Apply Muon to the invariant covectors The main idea of our approach is that we apply Muon's polar operation on H_U and H_V, which are *gauge invariant*. We then pull the resulting polar factors back with the same factors that created the covectors.

Definition 4.1 (Gauge-Muon(α))**.** *For learning rate $\eta > 0$, one exact Gauge-Muon step is*

$$U^+ = U - \eta A_\alpha \operatorname{polar}(H_U), \quad V^+ = V - \eta \operatorname{polar}(H_V) B_\alpha. \tag{6}$$

Theorem 4.2 (Exact finite-step gauge equivariance)**.** *Let $(U', V') = (DU, VD^{-1})$ for a positive diagonal D. If the two runs use the same learning rate and the same deterministic polar rule, then after one Gauge-Muon step*

$$(U'^+, V'^+) = (DU^+, V^+ D^{-1}). \tag{7}$$

By induction, gauge-equivalent initializations remain gauge equivalent for all subsequent steps as long as every active path has positive side norms.

In summary, Gauge-Muon works in three steps: (i) it first makes the covectors (H_U, H_V) gauge invariant, (ii) then applies Muon unchanged, (iii) and finally pulls the resulting direction back through the covariant factors. This differs from existing approaches to rescaling invariance. Path-SGD (Neyshabur et al., 2015a) obtains invariance through a path-normalized geometry, G-SGD (Meng et al., 2019) optimizes directly in an explicit parameterization of invariant path coordinates, and PathCond (Lebeurrier et al., 2026) rebalances the representative before training begins. In contrast, Gauge-Muon maintains Muon's matrix-level polar coupling. Thus, it cannot be viewed as spectral restatement of Path-SGD. Specifically, Path-SGD links each edge to others through the input-output paths they share, whereas in Gauge-Muon the singular vectors couple all units globally within each matrix block.

Extension to SwiGLU networks. Transformer feed-forward blocks typically use SwiGLU (Shazeer, 2020), $F(x) = V\,(\mathrm{SiLU}(W_g x) \odot (Ux))$ by having a gate projection W_g. Because SiLU is not positively homogeneous, the continuous symmetry is now $(W_g, U, V) \sim (W_g, DU, VD^{-1})$. Gauge-Muon therefore applies to (U, V) unchanged, with r_i, s_i, and p_i computed from U and V alone. For the gate, we use the invariant factor $\Omega_\alpha = \mathrm{diag}(p_i^{-\alpha})$ and update $W_g^+ = W_g - \eta\,\Omega_\alpha \operatorname{polar}(\Omega_\alpha G_g)$. This exponent is the only one under which the gate branch moves the function at the same path scale $p_i^{1-\alpha}$ as the up and down branches (cf. Proposition 6.1). The three-block update here is exactly equivariant under both the rescaling and hidden-unit permutations. Furthermore, it is steepest descent in a three-block case of the norm introduced in Section 5. Appendix E gives the detailed derivation of SwiGLU extension.

5 Gauge-Muon is quotient spectral steepest descent

The exact equivariance in Theorem 4.2 does not tell us how the optimization perform. This section will show the local geometry of Gauge-Muon update.

A state-dependent operator norm Consider a parameter perturbation $\Xi = (\Xi_U, \Xi_V)$ with the same shapes as (U, V). We introduce the following norm measuring the perturbation after removing the path-dependent pullback factors

$$\|\Xi\|_{\theta,\alpha} = \max\left\{ \left\|A_\alpha^{-1}\Xi_U\right\|_{\mathrm{op}}, \left\|\Xi_V B_\alpha^{-1}\right\|_{\mathrm{op}} \right\}, \quad \theta = (U, V). \tag{8}$$

To derive its dual, we denote any unit perturbation as $\Xi_U = A_\alpha Z_U$, $\Xi_V = Z_V B_\alpha$ with $\|Z_U\|_{\mathrm{op}}, \|Z_V\|_{\mathrm{op}} \le 1$. Then for a covector $G = (G_U, G_V)$,

$$\langle G, \Xi \rangle := \langle G_U, \Xi_U \rangle_F + \langle G_V, \Xi_V \rangle_F = \langle A_\alpha G_U, Z_U \rangle_F + \langle G_V B_\alpha, Z_V \rangle_F . \tag{9}$$

Applying operator–nuclear duality to the two blocks separately yields the dual norm

$$\|G\|_{\theta,\alpha,*} = \|A_\alpha G_U\|_* + \|G_V B_\alpha\|_* . \tag{10}$$

Theorem 5.1 (Variational characterization of Gauge-Muon)**.** *Let $S_\alpha(\theta) = (A_\alpha \operatorname{polar}(A_\alpha G_U), \operatorname{polar}(G_V B_\alpha) B_\alpha)$. Then $S_\alpha(\theta) \in \arg\max_{\|\Xi\|_{\theta,\alpha} \le 1} \langle \nabla \mathcal{L}(\theta), \Xi \rangle$, and the attained value is $\chi_\alpha(\theta) := \|A_\alpha G_U\|_* + \|G_V B_\alpha\|_*$. Consequently, the Gauge-Muon update, $\theta^+ = \theta - \eta S_\alpha(\theta)$ is exact steepest descent in the local norm in Equation 8.*

Why this is a quotient geometry. The tangent directions of the gauge orbit, $(\Lambda U, -V\Lambda)$ for diagonal Λ, change the parameterization but not the function, so the loss is flat along them and its gradient annihilates them. Consequently, maximizing the gradient's action over the unit ball of Equation 8 or over classes modulo these directions gives the same value χ_α, and Gauge-Muon is steepest descent on the quotient of parameter space by the rescaling group. Comparing to Muon (Bernstein & Newhouse, 2024b), only the unit ball changes and the polar oracle is the same but acts on invariant covectors. Unlike natural gradient (Amari, 1998), which targets all smooth reparameterizations with a Riemannian metric, Gauge-Muon removes one known architectural symmetry and uses a non-Riemannian max-of-operator-norms geometry. Appendix D gives the derivation.

6 What the exponent α controls

To this end, we have established the symmetry and geometry property of Gauge-Muon(α) for every α. That is, every exponent α is equivariant and quotient steepest descent. This section presents how different exponent α distribute motions across paths of different invariant strength.

6.1 First order functional mobility

We examine functional mobility which characterize how a hidden unit change the functional output. Consider the contribution of hidden unit i as $f_i(x) = v_i \,\mathrm{ReLU}(u_i^\top x)$. If we perturb the incoming and outgoing path (u_i, v_i) with $(\delta u_i, \delta v_i)$, the first-order perturbation is written as

$$\delta f_i(x) = \underbrace{\delta v_i \,\mathrm{ReLU}(u_i^\top x)}_{\text{outgoing branch}} + \underbrace{v_i \mathbf{1}_{u_i^\top x>0}\, \delta u_i^\top x}_{\text{incoming branch}}. \tag{11}$$

The norm of $\|\delta f_i(x)\|$ can be bounded as $\|\delta f_i(x)\| \le \|\delta v_i\|\,\|u_i\|\,\|x\| + \|v_i\|\,\|\delta u_i\| = (\|\delta v_i\|\, r_i + \|\delta u_i\|\, s_i)\,\|x\|$. Under Gauge-Muon step, we have the following proposition.

Proposition 6.1 (Pathwise first-order functional-mobility bound)**.** *Suppose one Gauge-Muon step with exponent α and learning rate η generate perturbation on path i, the first-order perturbation is bounded as*

$$\|\delta f_i(x)\|_2 \le 2\eta\,\|x\|_2\, p_i^{1-\alpha}. \tag{12}$$

This proposition says that up to factors independent of the path mass p_i, the largest first-order functional displacement available to a hidden path scales as $m_\alpha(p) := p^{\,1-\alpha}$. We call $m_\alpha(p)$ the *functional-mobility scale*. Its log-log elasticity with respect to path mass is $\frac{d\log m_\alpha(p)}{d\log p} = 1-\alpha$. This gives important interpretability for Gauge-Muon family. That is, for $\alpha < 1$, larger-mass paths have a larger first-order mobility scale. For $\alpha = 1$, the mobility scale is independent of path mass. For $\alpha > 1$, smaller-mass paths have a larger first-order mobility scale. Also, when $\alpha = 1/2$, Gauge-Muon has balanced canonical scalings $\sqrt{r_i/s_i}$ and $\sqrt{s_i/r_i}$.

Remark We have show that the exponent has an exact power-law interpretation in function space. Hence, we should not treat it as an optimizer parameters. While gauge invariance eliminates scaling symmetry in ReLU network, α decides how the optimization is distributed across invariant path strengths. Path-SGD (Neyshabur et al., 2015a) also normalizes updates using complementary path information, but it does not expose a matrix-spectral mobility family. Our $\alpha = 1$ case shares the intuition of complementary-side normalization, while the polar operation couples directions through singular vectors. PathCond (Lebeurrier et al., 2026) instead changes an orbit representative to improve conditioning before optimization.

Capacity and implicit geometry. The mobility law also has statistical and geometric consequences, developed in Appendix F.1. The total path mass $P(\theta) = \sum_i p_i$ is gauge invariant and bounds generalization; at $\alpha = 1$ every path receives the same capacity-growth allowance (Theorem F.1). In an idealized path flow, the exponent selects a mirror geometry that interpolates between entropic ($\alpha = 0$) and Euclidean ($\alpha = 1$), independently of the gauge, whereas the implicit geometry of Euclidean gradient flow depends on the gauge.

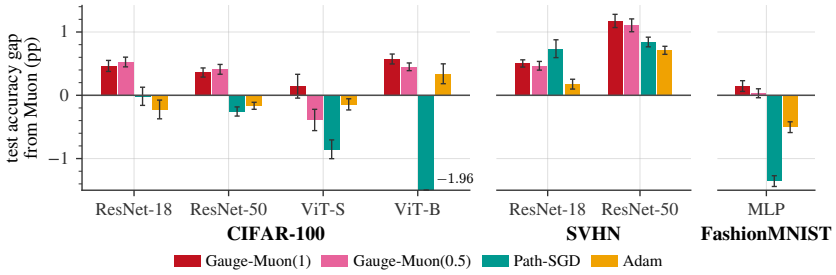


Figure 2: Test-accuracy gap against Muon (%), for each dataset and architecture. Bars are paired differences over matched seeds, with paired standard errors; positive values favor the plotted method. Bars below -1.5 are clipped and labelled with their value.

7 Experimental Results

This section studies the empirical behavior of Gauge-Muon. We first validate our theoretical predictions and then compare Gauge-Muon against established optimizers, including Muon (Jordan et al., 2024), Adam (Kingma & Ba, 2014), and Path-SGD (Neyshabur et al., 2015a).

7.1 Image classification task

Setup. We evaluate on FashionMNIST (Xiao et al., 2017), CIFAR-100 (Krizhevsky, 2009), and SVHN (Goodfellow et al., 2013), splitting each dataset into training, validation, and test sets. For each optimizer, we tune the learning rate on the validation split, retrain at the selected rate on five fresh seeds (three for the fine-tuned ViT), and report test accuracy over these seeds. The split sizes and learning-rate grids are given in Tables 2 and Table 3 of Appendix G.4.

Models. We consider 7 settings, each pairing a dataset with an architecture: a two-layer ReLU network on FashionMNIST; a frozen ImageNet-pretrained ResNet-18 or ResNet-50 backbone with a trainable two-layer ReLU classifier on CIFAR-100 and SVHN; and a ViT-S/16, ViT-B/16 (Dosovitskiy et al., 2020) fully fine-tuned on CIFAR-100. For the ViT, we use the checkpoint of You et al. (2024), whose activations are ReLU, so that the network is positively homogeneous as our analysis assumes. Appendix G.4.1 describes each architecture in detail.

Results. Figure 2 reports the test-accuracy gap of each optimizer relative to Muon. Gauge-Muon with $\alpha = 1$ improves on Muon in all settings, by up to 1.17 percentage points, and the gain exceeds one standard error in six of them. In contrast, Path-SGD falls below Muon in five of the seven settings and AdamW in four. Path-SGD is also gauge-equivariant, but it does not orthogonalize its update to have effective descent. Gauge-Muon combines both properties by keeping Muon orthogonalized update and applying it in a gauge-invariant geometry.

Further analyses can be found to the appendix including learning-rate sensitivity (Appendix G.4.2), validation-accuracy and training-loss curves (Appendix G.4.3), and robustness to the Newton–Schulz approximation and reduced precision (Appendix G.3).

7.2 Languge modeling task

Setup We use GPT-2 small settings (Radford et al., 2019) with 124M parameter to evaluate our models. We use FineWeb-Edu with two token budgets including 150M tokens and 1B tokens. We compare Gauge-Muon with three main baselines including Muon, Adam, and PathSGD. We search the best learning rate over a grid, report and compare best result according to the best learning rate. A detailed experiment setup is provided in Appendix G.

Results Figures 3 and 4 compare each optimizer with Muon, every optimizer at its own tuned learning rate. At 150M tokens, Gauge-Muon(1) beats Muon in all four settings, lowering the final

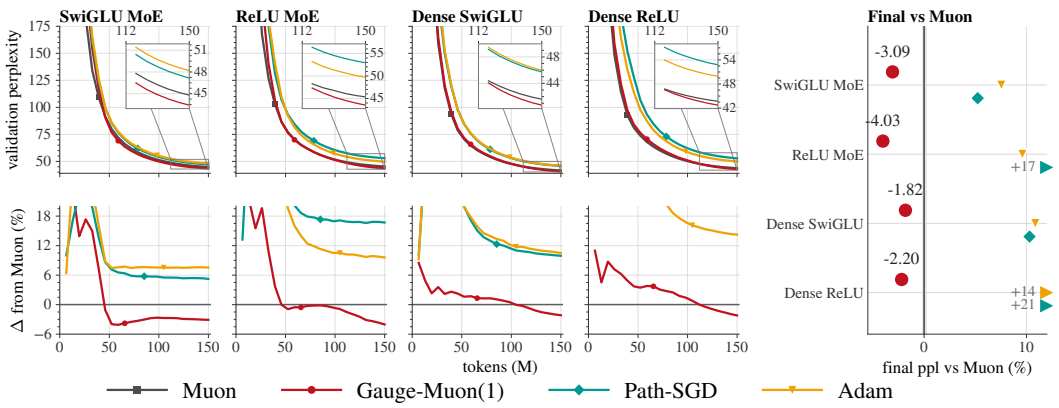


Figure 3: Training curves in the 150M-token setting. : validation perplexities, each at its tuned rate. The lower row is the difference from Muon at matched steps. *Right*: final validation perplexity relative to Muon. Large validation perplexity are clipped and labelled.

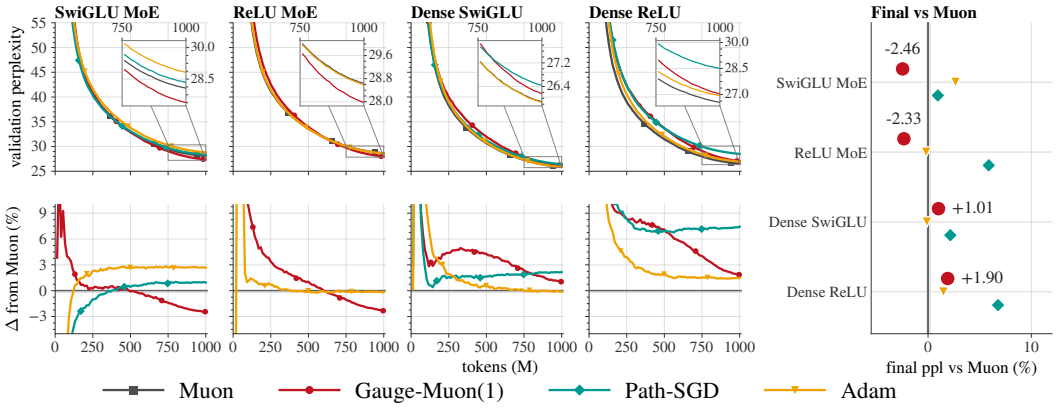


Figure 4: Training curves in the 1B-token setting.

validation perplexity by 1.8% to 4.0%. At 1B tokens, the result splits by architecture. In both mixture-of-experts models, Gauge-Muon(1) stays ahead of Muon (-2.5% and -2.3%), and the gap is still widening at the end of training. In both dense models, it falls behind we return to this case below.

The training curves share one pattern that Gauge-Muon(1) starts behind Muon, because its relative step ramps up from a small initialization, and then overtakes it in every setting except the dense models at 1B. The activation (SwiGLU or ReLU) changes the size of the difference but never its sign. Path-SGD, although also gauge-equivariant, trails Muon in every setting. Appendix G.5.3 gives the full results.

Width ablation Since the dense and mixture models differ mainly in the width of their feed-forward blocks, we vary the hidden width of the dense model to understand why Gauge-Muon trails Muon in the dense setting at 1B tokens (Figure 5, top). At 150M tokens, Gauge-Muon(1) leads Muon at every width. At 1B tokens, its advantage shrinks as the block widens and turns into a deficit once the width approaches the model dimension. Narrow blocks and short budgets therefore favor Gauge-Muon.

Mixture of experts Modern mixture-of-experts models use many experts that are each narrower than the model dimension, which is the regime the width ablation favors. We therefore vary the number of experts while holding the total feed-forward parameters fixed, so that more experts

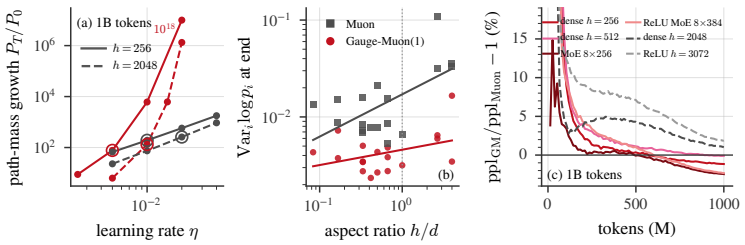


Figure 6: Why narrow blocks and short budgets favor Gauge-Muon. (a) Path-mass growth against the learning rate at 1B tokens; rings mark tuned rates. (b) Spread of log path mass at the end of training against the aspect ratio h/d, at tuned rates; lines are least-squares fits in log–log scale, with slope 0.44 for Muon and 0.16 for Gauge-Muon. (c) Perplexity gap to Muon over training at 1B tokens; solid lines are narrow blocks, dashed lines wide ones.

means narrower experts (Figure 5, bottom). The gain over Muon holds from 4 to 16 experts but vanishes at 32 experts of width 64, where the shared learning rate is likely too large for Gauge-Muon (Appendix G.5.5).

Why width and budget matter We investigate two properties that can explain this in Figure 6. First, Gauge-Muon moves every hidden unit by a fixed fraction of its own size, thereby path mass compounds and grows exponentially in the learning rate (see Theorem F.3). On the other hand, Muon steps have a fixed absolute size and its path mass grows only polynomially. Gauge-Muon's usable learning rate therefore has to shrink as the budget grows, which slows its early phase and delays the point where it overtakes Muon. Second, the aspect ratio of the block decides how Muon distributes its update. In a narrow block, Muon's orthogonalized update moves every hidden unit by the same amount. In a wide block, it moves units with larger gradients more and concentrates path mass on them, while Gauge-Muon keeps path mass evenly spread at every width (Figure 6 b). This concentration appears to help early in training: at 1B tokens every narrow block eventually overtakes Muon, while the wide blocks never do (Figure 6c). Appendix G.5.5 gives the details.

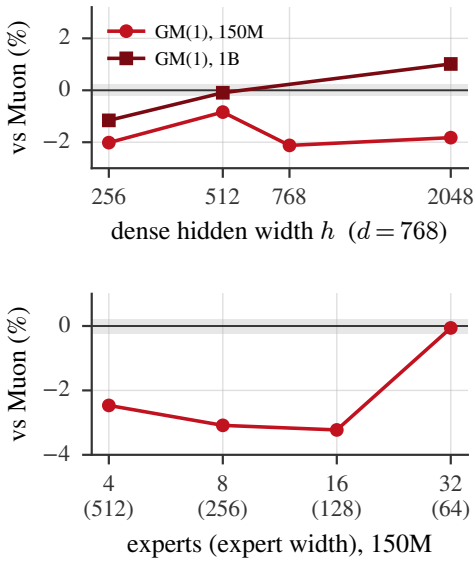


Figure 5: Perplexity relative to Muon at tuned rates. Top: dense hidden width. Bottom: expert count at fixed MLP parameters.

8 Conclusion

We introduced Gauge-Muon, a family of orthogonalized optimizers that respects the hidden-unit rescaling symmetry of ReLU and SwiGLU networks. It keeps Muon's polar map but applies it to gauge-invariant covectors, which makes every member of the family exactly equivariant and steepest descent on the quotient by the rescaling group. The family's exponent sets how functional motion is allocated across paths. Empirically, Gauge-Muon is unaffected by function-preserving rescalings and improves on Muon in image classification and in GPT-2 language modeling, most consistently with narrow feed-forward layers.

Limitations. Our language-modeling experiments are limited to small-scale models and token budgets. An open question that whether the gains persist at larger model and data scales is left for future work. We view the current study as a first characterization of Gauge-Muon rather than a scaling result.

Reproducibility Statement

Our source code for all experiments is provided as supplementary material. Full experimental settings, including model architectures, datasets, training procedures, and all hyperparameters, are provided in Appendix G.4 for image classification and Appendix G.5 for language modeling.

AI Use Statement

We used large language models (LLMs) to polish the writing, including editing for clarity and correcting grammar. We also used Claude Code to help refine our experimental pipeline and to run experiments under our direction. All research ideas, methods, and conclusions are our own.

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A NOTATIONS

Throughout the paper, we use the following notation. For a vector $x = (x_1, \dots, x_d)^\top \in \mathbb{R}^d$, its Euclidean norm is defined as $\|x\|_2 = \left(\sum_{j=1}^d x_j^2\right)^{1/2}$. For matrices $A, B \in \mathbb{R}^{m\times n}$, the Frobenius inner product is $\langle A, B\rangle_F = \mathrm{tr}(A^\top B)$, and the corresponding Frobenius norm is $\|A\|_F = \sqrt{\langle A, A\rangle_F}$. Every real matrix $A \in \mathbb{R}^{m\times n}$ admits a compact singular value decomposition (SVD) $A = P\Sigma Q^\top$, where $P \in \mathbb{R}^{m\times r}$ and $Q \in \mathbb{R}^{n\times r}$ have orthonormal columns, and $\Sigma = \mathrm{diag}(\sigma_1, \dots, \sigma_r)$ contains the positive singular values $\sigma_1 \geq \cdots \geq \sigma_r > 0$, with $r = \mathrm{rank}(A)$. The operator norm, also known as the spectral norm, measures the largest factor by which A can stretch a unit Euclidean vector and is defined as $\|A\|_{\mathrm{op}} = \max_{\|x\|_2\leq 1} \|Ax\|_2 = \sigma_1$. Its dual norm is the nuclear norm, defined as $\|A\|_* = \max_{\|B\|_{\mathrm{op}}\leq 1} \langle A, B\rangle_F = \sum_{j=1}^r \sigma_j$.

B EXTENDED RELATED WORK

We provide a broader related work in addition to Section 2 in the main text.

Spectral and orthogonalized optimizers. Beyond the original Muon (Jordan et al., 2024), subsequent work refined the polar iteration (Amsel et al., 2026), added adaptivity (Si et al., 2025), distributed the method (Ahn et al., 2025), and recast it within a unifying preconditioning view (Lau et al., 2025). Without accumulation, Shampoo's update reduces to the polar factor of the gradient (Bernstein & Newhouse, 2024b); SOAP (Vyas et al., 2025) is a further relative in the same family. The LMO view of norm-constrained training (Pethick et al., 2025) has roots in Frank-Wolfe (Jaggi, 2013). The spectral scaling condition (Yang et al., 2023) and μP (Yang & Hu, 2021; Yang et al., 2021) justify the operator-norm geometry through width-robust feature learning. MuonEq (Chang et al., 2026) also rescales the input to Muon before orthogonalization, but its row/column equilibration is computed from the current momentum matrix to improve finite-step Newton-Schulz conditioning. Gauge-Muon instead derives its paired rescaling from a function-preserving symmetry; its factors depend jointly on adjacent parameter blocks and guarantee equivariance across functionally equivalent representations.

Factor rescaling in low-rank adapters. Factor-rescaling non-invariance has also been identified for low-rank adapters, where $(B, A) \mapsto (BD, D^{-1}A)$ leaves BA unchanged. LoRA-RITE (Yen et al., 2025) constructs transformation-invariant adaptive preconditioners, and a Muon-type optimizer for parametrization-independent LoRA has been proposed (Bogachev et al., 2026). Our setting differs in that the gauge acts by a positive diagonal on a nonlinear (ReLU/SwiGLU) hidden layer rather than by $\mathrm{GL}(r)$ on a bilinear product.

Rescaling symmetry and path geometry. Rescaling invariance motivates path-norm capacity control (Neyshabur et al., 2015b; Golowich et al., 2018; Gonon et al., 2024). Besides Equinormalization (Stock et al., 2019), cross-layer equalization (Nagel et al., 2019) also selects a canonical orbit representative. Riemannian and quotient approaches treat the symmetry through the geometry of the quotient manifold (Absil et al., 2008; Boumal, 2020; Badrinarayanan et al., 2015; Cho & Lee, 2017). DDC (Shirodkar, 2026) lifts adaptive optimizers to architectural quotient spaces by separating symmetry and functional directions under an invariant metric; our question is narrower, namely which Muon-like spectral geometry should act on the quotient of the multiplicative hidden-unit gauge. Symmetry teleportation (Zhao et al., 2023a;b) instead moves along orbits to accelerate descent; see Zhao et al. (2025) for a survey. Scale symmetry also underlies reparameterization-dependent sharpness (Dinh et al., 2017) and its scale-invariant fixes (Kwon et al., 2021), as well as the effective-learning-rate dynamics of normalized layers (Salimans & Kingma, 2016; Hoffer et al., 2018; Arora et al., 2018). Gauge-Muon shares the orbit-invariance goal of Path-SGD and G-SGD, but, unlike canonicalization methods, it does not fix a single representative.

Reparameterization-invariant and natural-gradient methods. Natural gradient (Amari, 1998) and its approximations, such as K-FAC (Martens & Grosse, 2015) and unit-wise Riemannian metrics (Ollivier, 2013), are invariant to (classes of) reparameterizations through a Riemannian metric

on function or parameter space. Gauge-Muon achieves invariance with a *non-Riemannian*, state-dependent operator norm whose dual is a sum of nuclear norms of canonical covectors. Its update is therefore an LMO rather than a metric inverse, and it needs no curvature estimate.

Implicit bias. Beyond the mirror-descent characterizations cited in the main text (Gunasekar et al., 2018; Li et al., 2022), see also Gunasekar et al. (2020); the hyperbolic-entropy potentials of Ghai et al. (2019) are consistent with the entropic-to-Euclidean interpolation of Woodworth et al. (2019); Azulay et al. (2021). For homogeneous networks, steepest descent under different norms converges to different margin-maximizing solutions (Lyu & Li, 2019; Chizat & Bach, 2020; Tsilivis et al., 2024); related results characterize spectral descent and Muon (Chen et al., 2025). The capacity measure associated with our Box–Cox mirror flow is the gauge-invariant path/variation norm (Bach, 2014; Bach et al.).

Rescaling-aware quantization. Equivalence-preserving per-channel rescaling is a standard quantization tool. Cross-layer equalization (Nagel et al., 2019), SmoothQuant (Xiao et al., 2022), AWQ (Lin et al., 2023) and OmniQuant (Shao et al., 2023) all move scale between adjacent operators to reduce quantization error. We observe that per-row quantization is gauge-covariant, and we propose path-mass-based bit allocation, which, unlike magnitude heuristics, is gauge-invariant. We view this as a unifying lens rather than a new quantizer.

Other symmetry-aware matrix optimizers. Several recent works study symmetry and scale in matrix optimizers from complementary angles. Intrinsic Muon (Li et al., 2026) lifts unitarily invariant norm constraints through a Riemannian metric, which gives symmetry-preserving Muon updates on matrix manifolds such as fixed-rank, Stiefel, Grassmann, and SPD spaces; these manifolds are compact or unitarily invariant, whereas the hidden-unit rescaling quotient is noncompact and positive diagonal. Singh (2026) study gauge equivariance in factored matrix models under the orthogonal action $(U, V) \mapsto (UQ, VQ)$ and show that Muon respects it whereas coordinatewise adaptive methods do not. Orthogonal transformations are isometries of the spectral norm, whereas nonuniform positive diagonal rescalings are not, so ordinary Muon is equivariant to the orthogonal basis gauge but not to the multiplicative ReLU gauge. Xu et al. (2026) show that Muon's spectral geometry can have unfavorable width dependence and derive mean-normalized row/column geometries with improved smoothness and learning-rate transfer; width scaling is orthogonal to rescaling invariance. Nora (Yuan et al., 2026) projects row-wise momentum onto directions orthogonal to the current weights before row normalization, which suppresses radial jitter and gives an $\mathcal{O}(mn)$ Muon-like update respecting the single-weight radial symmetry $w \sim \lambda w$; the ReLU gauge $(u_i, v_i) \sim (d_i u_i, d_i^{-1} v_i)$ instead couples adjacent blocks. Closest in spirit is the unit-consistent adjoint framework (Uhlmann, 2026), which enforces diagonal-rescaling consistency at the level of the backward operator and addresses the same symmetry class. It asks how to make Euclidean descent consistent; we ask how to build a Muon-style spectral LMO after moving to gauge-consistent coordinates.

C Proofs

C.1 Proof of Theorem 4.2

Proof. Assume $r_i, s_i > 0$ for every i, so that A_α and B_α are well defined. At the transformed point $(U', V') = (DU, VD^{-1})$ we have $r'_i = d_i r_i$, $s'_i = d_i^{-1} s_i$, and $p'_i = p_i$, so all side norms remain positive and

$$A'_\alpha = \operatorname{diag}\big(d_i r_i p_i^{-\alpha}\big) = DA_\alpha, \qquad B'_\alpha = \operatorname{diag}\big(d_i^{-1} s_i p_i^{-\alpha}\big) = D^{-1} B_\alpha. \tag{13}$$

Since $\mathcal{L}(DU, VD^{-1}) = \mathcal{L}(U, V)$, the chain rule gives $G'_U = D^{-1} G_U$ and $G'_V = G_V D$ (Section 3.2). By Equation 5, the two runs therefore feed identical matrices $H'_U = H_U$ and $H'_V = H_V$ to the polar oracle. Because the polar rule is deterministic, both runs obtain the same $Q_U = \operatorname{polar}(H_U)$ and $Q_V = \operatorname{polar}(H_V)$. Substituting into Equation 6,

$$U'^{+} = DU - \eta (DA_\alpha) Q_U = D(U - \eta A_\alpha Q_U) = DU^{+}, \tag{14}$$

$$V'^{+} = VD^{-1} - \eta Q_V (D^{-1} B_\alpha) = (V - \eta Q_V B_\alpha) D^{-1} = V^{+} D^{-1}, \tag{15}$$

which is Equation 7.

For the finite-step statement, note that (U'^{+}, V'^{+}) and (U^{+}, V^{+}) are again related by the *same* gauge D. Hence, as long as all side norms of the current iterate are positive, the one-step argument applies again, and induction on the number of steps gives $(U'_t, V'_t) = (DU_t, V_t D^{-1})$ for every t. □

C.2 Proof of Theorem 5.1

Proof. Write $G = \nabla\mathcal{L}(\theta) = (G_U, G_V)$. Since A_α and B_α are invertible, the change of variables $Z_U = A_\alpha^{-1}\Xi_U$ and $Z_V = \Xi_V B_\alpha^{-1}$ is a bijection, and by Equation 8 it maps the unit ball $\{\|\Xi\|_{\theta,\alpha} \le 1\}$ onto the product set $\{\|Z_U\|_{\mathrm{op}} \le 1\} \times \{\|Z_V\|_{\mathrm{op}} \le 1\}$. Using $\langle X, DY\rangle_F = \langle DX, Y\rangle_F$ for diagonal D and the analogous identity for right multiplication,

$$\langle G, \Xi\rangle = \langle G_U, A_\alpha Z_U\rangle_F + \langle G_V, Z_V B_\alpha\rangle_F = \langle A_\alpha G_U, Z_U\rangle_F + \langle G_V B_\alpha, Z_V\rangle_F . \tag{16}$$

The constraint set is a product, so the two terms can be maximized independently. By Equation 1, the first term is maximized over $\|Z_U\|_{\mathrm{op}} \le 1$ by $Z_U = \mathrm{polar}(A_\alpha G_U)$, with value $\|A_\alpha G_U\|_*$, and the second by $Z_V = \mathrm{polar}(G_V B_\alpha)$, with value $\|G_V B_\alpha\|_*$. Mapping back, the maximizer is $\Xi = (A_\alpha \,\mathrm{polar}(A_\alpha G_U), \mathrm{polar}(G_V B_\alpha) B_\alpha) = S_\alpha(\theta)$, and the maximum equals $\chi_\alpha(\theta)$. Since $H_U = A_\alpha G_U$ and $H_V = G_V B_\alpha$, the step $\theta - \eta S_\alpha(\theta)$ coincides with Equation 6, which is therefore the steepest-descent step of length η in $\|\cdot\|_{\theta,\alpha}$. □

C.3 Proof of Proposition 6.1

Proof. Let δu_i and δv_i be the perturbations of u_i and v_i produced by one Gauge-Muon step, and write $Q_U = \mathrm{polar}(H_U)$ and $Q_V = \mathrm{polar}(H_V)$. Row i of $A_\alpha Q_U$ is $r_i p_i^{-\alpha}$ times row i of Q_U, and column i of $Q_V B_\alpha$ is $s_i p_i^{-\alpha}$ times column i of Q_V. Every row and every column of a matrix Q with $\|Q\|_{\mathrm{op}} \le 1$ has Euclidean norm at most one, since $\left\|Q^\top e_i\right\|_2 \le \|Q\|_{\mathrm{op}}$ and $\|Qe_i\|_2 \le \|Q\|_{\mathrm{op}}$. Because $\|\mathrm{polar}(H)\|_{\mathrm{op}} \le 1$, Equation 6 gives

$$\|\delta u_i\|_2 \le \eta r_i p_i^{-\alpha}, \qquad \|\delta v_i\|_2 \le \eta s_i p_i^{-\alpha}. \tag{17}$$

By Equation 11, the first-order variation of $f_i(x) = v_i \,\mathrm{ReLU}(u_i^\top x)$ is

$$\delta f_i(x) = \delta v_i \;\mathrm{ReLU}(u_i^\top x) + v_i \,\mathbf{1}_{\{u_i^\top x > 0\}}\, \delta u_i^\top x. \tag{18}$$

At the kink $u_i^\top x = 0$ the indicator may be replaced by any value in $[0,1]$ without affecting the argument below. By the triangle inequality,

$$\|\delta f_i(x)\|_2 \le \|\delta v_i\|_2 \left|\mathrm{ReLU}(u_i^\top x)\right| + \|v_i\|_2 \left|\delta u_i^\top x\right| . \tag{19}$$

Since $|\mathrm{ReLU}(z)| \le |z|$, the Cauchy–Schwarz inequality gives $\left|\mathrm{ReLU}(u_i^\top x)\right| \le r_i \|x\|_2$ and $\left|\delta u_i^\top x\right| \le \|\delta u_i\|_2 \|x\|_2$. Combining these bounds with Equation 17,

$$\|\delta f_i(x)\|_2 \le \eta s_i p_i^{-\alpha}\, r_i \|x\|_2 + s_i\, \eta r_i p_i^{-\alpha} \|x\|_2 = 2\eta \|x\|_2 \; r_i s_i \, p_i^{-\alpha} = 2\eta \|x\|_2 \; p_i^{1-\alpha}, \tag{20}$$

which is Equation 12. □

C.4 Exact one-path reduction

The mobility bound of Proposition 6.1 becomes an exact identity in the scalar model with a single hidden path. Let $u, v > 0$, $p = uv$, and suppose the loss is $\mathcal{L}(u, v) = \varphi(p)$ for a differentiable φ. Here $r = u$, $s = v$, $A_\alpha = up^{-\alpha}$, and $B_\alpha = vp^{-\alpha}$. The gradients are

$$\frac{\partial \mathcal{L}}{\partial u} = \varphi'(p) v, \qquad \frac{\partial \mathcal{L}}{\partial v} = \varphi'(p) u, \tag{21}$$

so both canonical covectors equal

$$H_u = H_v = p^{1-\alpha}\varphi'(p). \tag{22}$$

The polar factor of a scalar is its sign, with $\mathrm{polar}(0) = 0$ under the support convention. Writing $\kappa = \mathrm{sign}\,\varphi'(p) \in \{-1, 0, 1\}$ and defining the trust ratio $\rho = \eta p^{-\alpha}$, one Gauge-Muon step is

$$\boxed{\begin{aligned} u^+ &= u - \eta\, up^{-\alpha}\kappa = u\,(1 - \rho\kappa)\,, \\ v^+ &= v - \eta\,\kappa\, vp^{-\alpha} = v\,(1 - \rho\kappa)\,, \\ p^+ &= u^+ v^+ = p\,(1 - \rho\kappa)^2\,. \end{aligned}} \tag{23}$$

If $\rho < 1$, both factors stay positive, so the path cannot cross zero in one step. When $\varphi'(p) \neq 0$ we have $\kappa^2 = 1$, and expanding the last line gives

$$p^+ - p = -2\eta\kappa\, p^{1-\alpha} + \eta^2 p^{1-2\alpha}. \tag{24}$$

The first-order term has magnitude $2\eta p^{1-\alpha}$, matching the mobility scale $m_\alpha(p) = p^{1-\alpha}$, and the ratio of the second-order term to it is exactly

$$\frac{\eta^2 p^{1-2\alpha}}{2\eta p^{1-\alpha}} = \frac{\eta p^{-\alpha}}{2} = \frac{\rho}{2}. \tag{25}$$

Thus the trust ratio ρ that keeps the path positive also measures how accurate the first-order (continuous-time) mobility picture is.

D Quotient geometry of Gauge-Muon

This section gives the derivation behind the quotient statement in Section 5.

Vertical space. The *gauge orbit* through $\theta = (U, V)$ is $\{(DU, VD^{-1}) : D \in \mathcal{G}\}$. Writing $D(t) = \exp(t\Lambda)$ with Λ diagonal, its tangent directions are $\frac{d}{dt}(D(t)U, VD(t)^{-1})\big|_{t=0} = (\Lambda U, -V\Lambda)$, and we call

$$\mathcal{V}_\theta = \{(\Lambda U, -V\Lambda) : \Lambda \text{ diagonal}\} \tag{26}$$

the *vertical space*. A vertical vector is an infinitesimal change of parameterization, not of the represented function.

The loss differential annihilates vertical directions. Assume $\mathcal{L}(U, V)$ depends on the parameters only through the represented function. Since $f_{\theta(t)}$ is constant along the gauge curve, so is $\mathcal{L}(\theta(t))$, and by the chain rule

$$0 = \frac{d}{dt}\mathcal{L}(\theta(t))\bigg|_{t=0} = \langle \nabla_U \mathcal{L}, \Lambda U\rangle_F - \langle \nabla_V \mathcal{L}, V\Lambda\rangle_F = \langle \nabla\mathcal{L}(\theta), (\Lambda U, -V\Lambda)\rangle\,. \tag{27}$$

Hence $\langle \nabla\mathcal{L}(\theta), Z\rangle = 0$ for all $Z \in \mathcal{V}_\theta$: the loss is flat along the orbit, so its gradient is orthogonal to the orbit's tangent space.

Quotient norm. A tangent vector on the quotient is an equivalence class $[\Xi] = \Xi + \mathcal{V}_\theta$, with quotient norm

$$\|[\Xi]\|_{\theta/\mathcal{G},\alpha} = \inf_{Z \in \mathcal{V}_\theta} \|\Xi + Z\|_{\theta,\alpha}\,. \tag{28}$$

Because the loss differential vanishes on $\mathcal{V}_\theta$, maximizing its action over the ambient unit ball of Equation 8 or over quotient classes gives the same dual value $\chi_\alpha(\theta)$. The projection of $-S_\alpha$ is therefore a steepest-descent direction on the quotient space of functions modulo positive hidden-unit rescaling.

Remark D.1 (Comparison with vanilla Muon)**.** *Bernstein & Newhouse (2024b) view optimizers as steepest descent under architecture-dependent norms, and in this view Muon is steepest descent under the spectral norm. Theorem 5.1 identifies the state-dependent norm that makes this spectral geometry compatible with hidden-unit rescaling. Relative to Muon, only the unit ball changes, through the diagonal factors A_α, B_α; the polar oracle is the same but acts on the gauge-invariant covectors (H_U, H_V) rather than on the raw gradients, and the first-order decrease is measured by the invariant dual value χ_α instead of the nuclear norm of the raw gradient.*

Remark D.2 (Comparison with natural gradient)**.** *Natural gradient (Amari, 1998) replaces the Euclidean metric with the Fisher information and, in the ideal setting, is invariant to all smooth reparameterizations. Gauge-Muon instead quotients out only a known architectural symmetry and equips the remaining directions with the norm of Equation 8. That norm is a maximum of operator norms rather than a quadratic form, so the resulting geometry is symmetry-specific and non-Riemannian.*

E GAUGE-MUON FOR SWIGLU NETWORKS

Transformer feed-forward blocks commonly use a gated SwiGLU layer (Shazeer, 2020) rather than the ReLU layer of Equation 2. The SiLU activation (Elfwing et al., 2018), $\mathrm{SiLU}(z) = z/(1+e^{-z})$, is not positively homogeneous, so replacing ReLU by SiLU and keeping the ReLU gauge would be incorrect. Instead, we re-derive the symmetry from the SwiGLU computation itself and then extend Definition 4.1 one block at a time.

E.1 MODEL AND EXACT SYMMETRY

We keep the notation of Section 3.2 and add a gate projection. A SwiGLU block is

$$F(x) = V\,(\mathrm{SiLU}(W_g x) \odot (Ux)) = \sum_{i=1}^{h} v_i\ \mathrm{SiLU}(g_i^\top x)\,(u_i^\top x), \tag{29}$$

where $U \in \mathbb{R}^{h\times d}$ is the up projection (incoming matrix), $V \in \mathbb{R}^{k\times h}$ is the down projection (outgoing matrix), $W_g \in \mathbb{R}^{h\times d}$ is the gate projection with rows $g_i^\top$, and $\odot$ denotes elementwise multiplication. As in Equation 3, we set $r_i = \|u_i\|_2$, $s_i = \|v_i\|_2$, and $p_i = r_i s_i$. The gate norm $\|g_i\|_2$ does not enter p_i: because SiLU is not homogeneous, the gate scale generally changes the represented function.

Continuous symmetry. For a positive diagonal D, the map

$$(W_g, U, V) \mapsto (W_g,\ DU,\ VD^{-1}) \tag{30}$$

leaves F unchanged, since $\mathrm{SiLU}(W_g x)\odot(DUx) = D\,(\mathrm{SiLU}(W_g x)\odot(Ux))$ and D cancels against D^{-1} in VD^{-1}. The gate admits no such rescaling, because $\mathrm{SiLU}(Dz) \neq D\,\mathrm{SiLU}(z)$ in general. The ReLU gauge therefore survives only on the linear up branch and the down projection, and the path mass p_i remains an exact invariant.

Permutation symmetry. For any permutation matrix P, the map

$$(W_g, U, V) \mapsto (PW_g,\ PU,\ VP^\top) \tag{31}$$

also preserves F. Together with Equation 30, it generates a monomial action on the hidden units. The same permutation symmetry holds for the ReLU layer.

E.2 CHOOSING THE GATE FACTOR

For the up and down blocks, we reuse the covariant factors of Section 4, $A_\alpha = \mathrm{diag}(r_i p_i^{-\alpha})$ and $B_\alpha = \mathrm{diag}(s_i p_i^{-\alpha})$. The gate is not rescaled by the gauge, so its factor must itself be gauge invariant. We therefore take $\Omega_\gamma = \mathrm{diag}(p_i^{-\gamma})$ for some exponent γ and fix γ by requiring that all three blocks move the function at the same path scale. This extends Proposition 6.1.

The first-order variation of the contribution of unit i, $f_i(x) = v_i\ \mathrm{SiLU}(g_i^\top x)(u_i^\top x)$, is

$$\delta f_i(x) = \underbrace{\delta v_i\ \mathrm{SiLU}(g_i^\top x)(u_i^\top x)}_{\text{down}} + \underbrace{v_i\ \mathrm{SiLU}(g_i^\top x)(\delta u_i^\top x)}_{\text{up}} + \underbrace{v_i\ \mathrm{SiLU}'(g_i^\top x)(u_i^\top x)(\delta g_i^\top x)}_{\text{gate}}. \tag{32}$$

Proposition E.1 (Three-branch functional mobility)**.** *Suppose one step moves the up and down blocks by $\eta A_\alpha Q_U$ and $\eta Q_V B_\alpha$ and the gate by $\eta\Omega_\gamma Q_g$, where $\|Q_U\|_{\mathrm{op}}, \|Q_V\|_{\mathrm{op}}, \|Q_g\|_{\mathrm{op}} \le 1$. Let $\kappa = \sup_z |\,\mathrm{SiLU}'(z)| < 1.1$. Then*

$$\|\delta f_i(x)\|_2 \le 2\eta\ \|x\|_2\ |\,\mathrm{SiLU}(g_i^\top x)|\ p_i^{1-\alpha} + \eta\,\kappa\ \|x\|_2^2\ p_i^{1-\gamma}. \tag{33}$$

The three branches share the same path-mass exponent for every $p_i > 0$ if and only if $\gamma = \alpha$.

Proof. As in the proof of Proposition 6.1, a matrix with unit operator norm has rows and columns of Euclidean norm at most one. Hence $\|\delta u_i\|_2 \le \eta r_i p_i^{-\alpha}$, $\|\delta v_i\|_2 \le \eta s_i p_i^{-\alpha}$, and $\|\delta g_i\|_2 \le \eta p_i^{-\gamma}$. By Cauchy–Schwarz, the down branch is bounded by $\eta s_i p_i^{-\alpha}\,|\,\mathrm{SiLU}(g_i^\top x)|\,r_i\,\|x\|_2$ and the up branch by $s_i\,|\,\mathrm{SiLU}(g_i^\top x)|\ \|x\|_2\ \eta r_i p_i^{-\alpha}$. Each equals $\eta\,\|x\|_2\,|\,\mathrm{SiLU}(g_i^\top x)|\,p_i^{1-\alpha}$. The gate branch is bounded by $s_i\,\kappa\,r_i\,\|x\|_2\ \|x\|_2\ \eta p_i^{-\gamma} = \eta\kappa\,\|x\|_2^2\ p_i^{1-\gamma}$. Summing the three bounds gives Equation 33. The exponents $1-\alpha$ and $1-\gamma$ agree for every $p_i > 0$ exactly when $\gamma = \alpha$. □

The gate factor is therefore derived rather than chosen for symmetry: with $\gamma = \alpha$, the gate is not gauge-equivalent to the other blocks, but its update mobility is matched to the same invariant path scale $p_i^{1-\alpha}$. We write $\Omega_\alpha = \mathrm{diag}(p_i^{-\alpha})$ from now on.

E.3 The update and its guarantees

Let G_g, G_U, G_V denote the gradients with respect to W_g, U, V. The canonical covectors are

$$H_g = \Omega_\alpha G_g, \qquad H_U = A_\alpha G_U, \qquad H_V = G_V B_\alpha, \tag{34}$$

and one Gauge-Muon step for SwiGLU is

$$\begin{aligned} W_g^+ &= W_g - \eta\, \Omega_\alpha\, \mathrm{polar}(H_g), \\ U^+ &= U - \eta\, A_\alpha\, \mathrm{polar}(H_U), \\ V^+ &= V - \eta\, \mathrm{polar}(H_V) B_\alpha. \end{aligned} \tag{35}$$

The up and down updates are exactly those of Definition 4.1. The gate adds a third block whose factor is invariant.

Theorem E.2 (Exact monomial equivariance)**.** *Assume the same deterministic polar rule is used in both runs and that it commutes with row and column permutations, as the support polar does. Then the update in Equation 35 commutes exactly with the rescaling in Equation 30 and the permutation in Equation 31. In particular, gauge-equivalent or permuted initializations produce the same function-space trajectory in exact arithmetic.*

Proof. Under Equation 30, p_i is unchanged, so $\Omega'_\alpha = \Omega_\alpha$, while $A'_\alpha = DA_\alpha$ and $B'_\alpha = D^{-1}B_\alpha$ as in the ReLU case. By the chain rule, $G'_g = G_g$, $G'_U = D^{-1}G_U$, and $G'_V = G_V D$, so all three covectors in Equation 34 are invariant. The gate update is then identical in both runs, and the up and down updates transform as in the proof of Theorem 4.2.

Under Equation 31, the path masses are permuted, so $\Omega'_\alpha = P\Omega_\alpha P^\top$, $A'_\alpha = PA_\alpha P^\top$, and $B'_\alpha = PB_\alpha P^\top$. The gradients transform as $G'_g = PG_g$, $G'_U = PG_U$, and $G'_V = G_V P^\top$, which gives $H'_g = PH_g$, $H'_U = PH_U$, and $H'_V = H_V P^\top$. Since $\mathrm{polar}(PH) = P\,\mathrm{polar}(H)$ and $\mathrm{polar}(HP^\top) = \mathrm{polar}(H)P^\top$, each pulled-back update permutes exactly as its parameter block. □

Quotient steepest descent. The argument of Section 5 carries over block by block. The local norm

$$\|\Xi\|_{\mathrm{sw},\theta,\alpha} = \max\left\{ \left\|\Omega_\alpha^{-1}\Xi_g\right\|_{\mathrm{op}}, \left\|A_\alpha^{-1}\Xi_U\right\|_{\mathrm{op}}, \left\|\Xi_V B_\alpha^{-1}\right\|_{\mathrm{op}} \right\} \tag{36}$$

has dual

$$\|G\|_{\mathrm{sw},\theta,\alpha,*} = \|\Omega_\alpha G_g\|_* + \|A_\alpha G_U\|_* + \|G_V B_\alpha\|_*, \tag{37}$$

and, by the proof of Theorem 5.1 applied to each of the three blocks, Equation 35 is exact steepest descent in the norm of Equation 36. The vertical space is now $\mathcal{V}_\theta = \{(0, \Lambda U, -V\Lambda) : \Lambda \text{ diagonal}\}$, which has no gate component because the continuous gauge fixes W_g. The loss gradient is orthogonal to $\mathcal{V}_\theta$ by the same argument as in Section 5, so the step is also steepest descent on the quotient by positive up–down rescaling.

Relation to other symmetry-aware optimizers. Recent symmetry-compatible optimizers derive block-specific updates from permutation and other architectural symmetries (Lau & Su, 2026). The continuous up–down quotient used here is complementary: permutations reorder hidden units, whereas positive diagonal scaling changes how each unit is factored between U and V without changing the block function.

F Path capacity and implicit geometry

F.1 Overview

The mobility law of Proposition 6.1 also has statistical and geometric consequences. By positive homogeneity, $f_\theta(x) = \sum_i (r_i v_i)\,\mathrm{ReLU}(\hat{u}_i^\top x)$ with unit-norm features $\hat{u}_i = u_i/r_i$, so the total path

mass $P(\theta) = \sum_i p_i$ is the sum of the Euclidean norms of the output coefficients. It is gauge invariant and certifies the capacity of the current representation, independently of how each unit is balanced between U and V.

Theorem F.1 (Invariant capacity budget, informal)**.** *Let $\|x\|_2 \le R$ and let the loss be G-Lipschitz in the prediction $f(x) \in \mathbb{R}^k$. Uniformly over networks with $P(\theta) \le D$, the expected generalization gap is at most $8kGDR/\sqrt{n}$. Moreover, if a T-step Gauge-Muon run with $\alpha = 1$, whose horizon and initialization are fixed independently of the sample, satisfies the trust bound $\eta p_{i,t}^{-1} \le \rho_0$, then*

$$P(\theta_T) \le P(\theta_0) + (2+\rho_0)\eta h T, \qquad \mathbb{E}\big[\mathcal{R}(f_{\theta_T}) - \widehat{\mathcal{R}}_S(f_{\theta_T})\big] \le \frac{8kGR}{\sqrt{n}}\big[P(\theta_0) + (2+\rho_0)\eta h T\big]. \quad (38)$$

The exponent that equalizes first-order functional mobility therefore also gives every active path the same worst-case capacity-growth allowance, and early stopping acquires a direct meaning as control of an invariant capacity budget. The guarantee is conservative: the statistical bound is width independent at fixed P, but the trajectory-based budget grows with hT, and neither claims that $\alpha = 1$ always generalizes best.

The exponent also selects the geometry in which paths compete. With frozen feature directions and the polar demand idealized by the loss gradient, the reduced dynamics $\dot{p}_i = -p_i^{1-\alpha}\partial_i F(p)$ are exactly mirror flow under a Box–Cox potential with curvature $p^{\alpha-1}$. The range $0 \le \alpha \le 1$ interpolates between entropic ($\alpha = 0$) and Euclidean ($\alpha = 1$) path geometry, and, conditional on positivity and convergence, the limiting interpolator is the Bregman projection of the initialization. Euclidean gradient flow, in contrast, induces a path mobility that depends on the gauge-dependent imbalance $\|u_i\|_2^2 - \|v_i\|_2^2$, which a function-preserving rescaling can distort without bound. Gauge-Muon thus makes both the capacity certificate and the reduced implicit geometry properties of the invariant paths. The rest of this section gives the precise statements, proofs, and limitations.

This section connects the invariant path coordinates of Gauge-Muon to statistical capacity and optimization geometry. We first bound generalization by the total path mass and control its growth under finite Gauge-Muon steps. We then characterize the mirror geometry of an idealized path flow and quantify how Euclidean gradient flow depends on the choice of gauge.

Throughout, we consider the ReLU layer of Equation 2 with vector output, $f_\theta(x) = \sum_{i=1}^h v_i \,\mathrm{ReLU}(u_i^\top x) \in \mathbb{R}^k$, and the magnitudes of Equation 3: $r_i = \|u_i\|_2$, $s_i = \|v_i\|_2$, and $p_i = r_i s_i$. The scalar-output case is $k = 1$, where $s_i = |v_i|$. The total path mass is

$$P(\theta) := \sum_{i=1}^h p_i = \sum_{i=1}^h \|u_i\|_2 \, \|v_i\|_2 \,. \quad (39)$$

F.2 Path mass as a capacity certificate

Path normalization. According to positive homogeneity of ReLU, we have

$$f_\theta(x) = \sum_i a_i \,\mathrm{ReLU}(\hat{u}_i^\top x), \qquad \hat{u}_i = u_i/r_i, \quad a_i = r_i v_i \in \mathbb{R}^k, \qquad P(\theta) = \sum_i \|a_i\|_2 \,. \quad (40)$$

We omit any units with $r_i = 0$. We define the variation norm

$$\gamma_1(f) := \inf\left\{ \sum_j \|a_j\|_2 : f = \sum_j a_j \,\mathrm{ReLU}(w_j^\top \cdot),\ a_j \in \mathbb{R}^k,\ \|w_j\|_2 = 1 \right\}. \quad (41)$$

Equation 40 certifies $\gamma_1(f_\theta) \le P(\theta)$, and P is invariant under $(u_i, v_i) \mapsto (d_i u_i, d_i^{-1} v_i)$ for $d_i > 0$. The certificate refers to the current representation: distinct representations of the same function can have different total path mass.

Setting. Let $\mathcal{F}_D$ contain all finite sums $\sum_j a_j \,\mathrm{ReLU}(w_j^\top \cdot)$ with $\|w_j\|_2 = 1$ and $\sum_j \|a_j\|_2 \le D$, for a fixed D. For an i.i.d. sample $S = \{(x_j, y_j)\}_{j=1}^n$, let $\mathcal{R}(f) = \mathbb{E}\,\ell(y, f(x))$ and $\widehat{\mathcal{R}}_S(f) = n^{-1}\sum_j \ell(y_j, f(x_j))$. Assume $\|x\|_2 \le R$ almost surely, $\ell(y, \cdot)$ is G-Lipschitz with respect to the Euclidean norm on $\mathbb{R}^k$, and $\mathbb{E}|\ell(y, 0)| < \infty$.

Theorem F.2 (Generalization under a path-mass budget)**.** *Under these assumptions,*

$$\mathfrak{R}_n(\ell \circ \mathcal{F}_D) \le \frac{4kGDR}{\sqrt{n}}, \qquad \mathbb{E}_S\left[\sup_{f\in\mathcal{F}_D}\left(\mathcal{R}(f) - \widehat{\mathcal{R}}_S(f)\right)\right] \le \frac{8kGDR}{\sqrt{n}}. \tag{42}$$

An empirical risk minimizer $\widehat{f}$ over $\mathcal{F}_D$, when it exists, satisfies $\mathbb{E}\,\mathcal{R}(\widehat{f}) - \inf_{f\in\mathcal{F}_D}\mathcal{R}(f) \le 8kGDR/\sqrt{n}$.

Proof. Let σ_j be independent Rademacher signs and σ_{jc}, $c = 1, \dots, k$, independent Rademacher signs arranged as vectors $\boldsymbol{\sigma}_j \in \{\pm 1\}^k$. Subtracting $\ell(y_j, 0)$ does not change the Rademacher complexity, so we may assume $\ell(y_j, 0) = 0$.

By the vector-contraction inequality (Maurer, 2016),

$$\mathbb{E}_\sigma \sup_{f\in\mathcal{F}_D}\sum_j \sigma_j \ell(y_j, f(x_j)) \le \sqrt{2}\,G\,\mathbb{E}_\sigma \sup_{f\in\mathcal{F}_D}\sum_j \langle \boldsymbol{\sigma}_j, f(x_j)\rangle\,.$$

For $f = \sum_i a_i \,\mathrm{ReLU}(w_i^\top \cdot) \in \mathcal{F}_D$, we have

$$\begin{aligned}
\sum_j \langle \boldsymbol{\sigma}_j, f(x_j)\rangle &= \sum_i \Big\langle a_i, \sum_j \boldsymbol{\sigma}_j \,\mathrm{ReLU}(w_i^\top x_j)\Big\rangle \\
&\le D \sup_{\|w\|_2=1}\Big\|\sum_j \boldsymbol{\sigma}_j \,\mathrm{ReLU}(w^\top x_j)\Big\|_2 \\
&\le D\sum_{c=1}^{k}\sup_{\|w\|_2=1}\Big|\sum_j \sigma_{jc}\,\mathrm{ReLU}(w^\top x_j)\Big|,
\end{aligned}$$

where the last step uses $\|z\|_2 \le \|z\|_1$.

Since ReLU is 1-Lipschitz with $\mathrm{ReLU}(0) = 0$, the contraction principle (Ledoux & Talagrand, 1991) and Jensen's inequality give, for each c,

$$\mathbb{E}_\sigma \sup_{\|w\|_2=1}\Big|\sum_j \sigma_{jc}\,\mathrm{ReLU}(w^\top x_j)\Big| \le 2\,\mathbb{E}_\sigma\Big\|\sum_j \sigma_{jc} x_j\Big\|_2 \le 2\Big(\sum_j \|x_j\|_2^2\Big)^{1/2} \le 2R\sqrt{n}.$$

Combining the three steps and dividing by n gives $\mathfrak{R}_n(\ell\circ\mathcal{F}_D) \le 2\sqrt{2}\,kGDR/\sqrt{n} \le 4kGDR/\sqrt{n}$. Symmetrization doubles this and yields the uniform bound. For any fixed $f \in \mathcal{F}_D$, empirical optimality of $\widehat{f}$ gives

$$\mathcal{R}(\widehat{f}) - \mathcal{R}(f) \le \sup_{g\in\mathcal{F}_D}\left[\mathcal{R}(g) - \widehat{\mathcal{R}}_S(g)\right] + \widehat{\mathcal{R}}_S(f) - \mathcal{R}(f).$$

The last term has zero expectation. Taking expectations and then the infimum over f proves the estimation bound. □

At fixed D the bound does not depend on the width h. The factor k comes from the crude step $\|z\|_2 \le \|z\|_1$ and is not optimized. By Theorem 4.2, gauge-equivalent Gauge-Muon runs have identical $P(\theta_t)$, so their capacity certificates agree throughout training. Equivariance alone, however, does not imply a smaller certificate or a smaller population risk.

F.3 Finite-step capacity growth

We use the exact update of Definition 4.1, without additional blockwise multipliers or damping. Let q_i^U denote row i of $\mathrm{polar}(H_U)$ and q_i^V column i of $\mathrm{polar}(H_V)$. Both have Euclidean norm at most one, and

$$\Delta u_i = -\eta r_i p_i^{-\alpha} q_i^U, \qquad \Delta v_i = -\eta s_i p_i^{-\alpha} q_i^V, \qquad \rho_i := \eta p_i^{-\alpha}. \tag{43}$$

The bounds below assume positive path masses at every pre-update state.

Theorem F.3 (Capacity budget of Gauge-Muon). *For T steps with fixed α and η, suppose $\rho_{i,t} \le \rho_0$ for every unit and step. Then*

$$P(\theta_T) \le P(\theta_0) + (2+\rho_0)\eta \sum_{t=0}^{T-1} \sum_{i=1}^{h} p_{i,t}^{1-\alpha}. \tag{44}$$

In particular, at $\alpha = 1$,

$$P(\theta_T) \le D_T := P(\theta_0) + (2+\rho_0)\eta h T. \tag{45}$$

Proof. The triangle inequality gives $r_i^+ \le r_i(1+\rho_i)$ and $s_i^+ \le s_i(1+\rho_i)$. Hence

$$p_i^+ - p_i \le p_i(2\rho_i + \rho_i^2) = (2+\rho_i)\eta p_i^{1-\alpha}.$$

Summing over units and steps and using $\rho_i \le \rho_0$ gives Equation 44. At $\alpha = 1$, each positive path contributes one to the inner sum. □

If $P(\theta_0)$, η, T, and ρ_0 are fixed independently of the training sample and the trust bound holds almost surely, then $f_{\theta_T} \in \mathcal{F}_{D_T}$ and Theorem F.2 gives, at $\alpha = 1$,

$$\mathbb{E}\big[\mathcal{R}(f_{\theta_T}) - \widehat{\mathcal{R}}_S(f_{\theta_T})\big] \le \frac{8kGR}{\sqrt{n}}\big[P(\theta_0) + (2+\rho_0)\eta h T\big]. \tag{46}$$

An independent random initialization can instead be conditioned on and then averaged over. Neither empirical optimality nor convexity of the loss is needed. A predetermined training horizon therefore controls a worst-case capacity budget. The statistical bound is width independent at fixed path mass, whereas this optimization-based certificate grows with hT.

For $\alpha < 1$, the growth allowance $p_i^{1-\alpha}$ increases with path mass; for $\alpha > 1$, it increases as paths weaken; at $\alpha = 1$, it is uniform across paths. These are upper bounds on capacity growth, not claims about realized growth or about an optimal exponent. The trust condition controls the quadratic correction and must be verified or enforced; it does not follow from the choice of α.

F.4 Reduced path dynamics and Box–Cox geometry

Lemma F.4 (Realized path demand). *At a fixed state with $p_i > 0$, the update in Equation 43 satisfies, as $\eta \to 0$,*

$$p_i^+ - p_i = \eta \nu_i p_i^{1-\alpha} + O(\eta^2 p_i^{1-2\alpha}), \qquad \nu_i = -\big(\langle \hat{u}_i, q_i^U \rangle + \langle \hat{v}_i, q_i^V \rangle\big), \quad |\nu_i| \le 2, \tag{47}$$

where $\hat{u}_i = u_i/r_i$ and $\hat{v}_i = v_i/s_i$.

Proof. Differentiating the norms gives $\Delta r_i = \langle \hat{u}_i, \Delta u_i \rangle + O(r_i\rho_i^2)$ and $\Delta s_i = \langle \hat{v}_i, \Delta v_i \rangle + O(s_i\rho_i^2)$. Expanding $(r_i + \Delta r_i)(s_i + \Delta s_i)$ and substituting Equation 43 gives the expansion. Cauchy–Schwarz and the norm bounds on q_i^U, q_i^V give $|\nu_i| \le 2$. □

The factor $p_i^{1-\alpha}$ sets the mobility, while ν_i records the demand realized by the matrix polar and is coupled across units. To isolate mobility, we freeze the signed features $\phi_i(x) = \hat{v}_i \,\mathrm{ReLU}(\hat{u}_i^\top x) \in \mathbb{R}^k$ and consider $f_p(x) = \sum_i p_i \phi_i(x)$ with empirical risk $F(p)$. Replacing the polar demand by $-\partial_i F(p)$ defines the idealized flow

$$\dot{p}_i = -p_i^{1-\alpha} \partial_i F(p). \tag{48}$$

This replacement is a modeling assumption: the full optimizer also changes feature directions and generally has $\nu_i \ne -\partial_i F(p)$.

Theorem F.5 (Box–Cox mirror flow). *Let $\alpha \ge 0$ and let F be differentiable and convex on $\mathbb{R}^h_{>0}$. Define $\Phi_\alpha(p) = \sum_i \varphi_\alpha(p_i)$ with*

$$\varphi_\alpha(z) = \begin{cases} \dfrac{z^{\alpha+1}}{\alpha(\alpha+1)} - \dfrac{z}{\alpha}, & \alpha > 0, \\ z\log z - z, & \alpha = 0. \end{cases} \tag{49}$$

While the trajectory remains positive, Equation 48 is exactly the mirror flow

$$\frac{d}{dt}\nabla\Phi_\alpha(p(t)) = -\nabla F(p(t)). \tag{50}$$

Suppose further that $F(p) = \mathcal{L}(\Psi p)$ for a linear map $\Psi : \mathbb{R}^h \to \mathbb{R}^{nk}$ and differentiable $\mathcal{L}$, and that the trajectory converges to a positive interpolator $p_\infty \in \mathcal{M} := \{p > 0 : \Psi p = y\}$. Then

$$p_\infty = \arg\min_{p\in\mathcal{M}} D_{\Phi_\alpha}(p, p_0), \qquad D_\Phi(p, q) := \Phi(p) - \Phi(q) - \langle \nabla\Phi(q), p - q\rangle . \tag{51}$$

Proof. Since $\varphi''_\alpha(z) = z^{\alpha-1} > 0$, the chain rule gives Equation 50, and Φ_α is strictly convex. Moreover, $\nabla F(p) \in \mathrm{range}(\Psi^\top)$, so integrating and using that this finite-dimensional subspace is closed gives $\nabla\Phi_\alpha(p_\infty) - \nabla\Phi_\alpha(p_0) \in \mathrm{range}(\Psi^\top)$. This is the first-order optimality condition for minimizing $D_{\Phi_\alpha}(\cdot, p_0)$ subject to $\Psi p = y$. Feasibility and strict convexity make the minimizer unique. □

The dual coordinate is the Box–Cox transform $\varphi'_\alpha(z) = (z^\alpha - 1)/\alpha$, with limit $\log z$ at $\alpha = 0$. In particular,

$$D_{\Phi_1}(p, p_0) = \tfrac{1}{2}\left\|p - p_0\right\|_2^2, \qquad D_{\Phi_0}(p, p_0) = \sum_i \left[p_i \log\frac{p_i}{p_{0,i}} - p_i + p_{0,i}\right], \tag{52}$$

so $0 \le \alpha \le 1$ interpolates between entropic and Euclidean path geometry. For $p_0 = \varepsilon\mathbf{1}$,

$$D_{\Phi_0}(p, \varepsilon\mathbf{1}) = \log(1/\varepsilon)\sum_i p_i + \sum_i (p_i \log p_i - p_i) + h\varepsilon.$$

After division by $\log(1/\varepsilon)$, this converges to the total path mass for each fixed p. On a fixed compact feasible subset of the positive orthant the convergence is uniform, so accumulation points of minimizers minimize total path mass. Pointwise convergence alone does not establish convergence of minimizers on the unrestricted interpolation set.

The theorem is conditional on positivity and convergence, neither of which is proved here; for $\alpha > 0$ the flow can reach the boundary. The Bregman projection characterizes the reduced gradient-demand model, not the implicit bias of the full spectrally coupled optimizer.

F.5 Gauge dependence of Euclidean gradient flow

For comparison, we keep the same frozen features and optimize the positive radial parameters r_i, s_i by Euclidean gradient flow on $F(r \odot s)$. The comparison holds while these parameters remain positive.

Proposition F.6 (Representative-dependent path geometry)**.** *The induced dynamics satisfy*

$$\dot r_i = -s_i\partial_i F(p), \qquad \dot s_i = -r_i\partial_i F(p), \qquad \dot p_i = -\sqrt{4p_i^2 + c_i^2}\,\partial_i F(p), \tag{53}$$

where the imbalance $c_i = r_i^2 - s_i^2$ is conserved. The rescaling $(r_i, s_i) \mapsto (d_i r_i, d_i^{-1} s_i)$ preserves p_i but changes c_i to $d_i^2 r_i^2 - d_i^{-2} s_i^2$, and hence generally changes the mobility.

Proof. The first two identities follow from the chain rule. They imply $\dot c_i = 2r_i\dot r_i - 2s_i\dot s_i = 0$ and $\dot p_i = -(r_i^2 + s_i^2)\partial_i F(p)$. Finally, $(r_i^2 + s_i^2)^2 = 4p_i^2 + c_i^2$. □

Relative to the balanced representative, the mobility is multiplied by

$$\frac{\sqrt{4p_i^2 + c_i^2}}{2p_i} = \frac{1}{2}\left(\frac{r_i}{s_i} + \frac{s_i}{r_i}\right) = \cosh\left(\log\frac{r_i}{s_i}\right). \tag{54}$$

This factor is minimized at balance and unbounded along a gauge orbit. Since it can differ across units, a rescaling can change relative path speeds, not just the global time scale.

Equivalently, Euclidean radial flow is a mirror flow with a separable potential of curvature $\psi''_{c_i}(p) = (4p^2 + c_i^2)^{-1/2}$ and dual coordinate

$$\psi'_{c_i}(p) = \begin{cases} \frac{1}{2}\operatorname{arsinh}(2p/|c_i|), & c_i \neq 0, \\ \frac{1}{2}\log p, & c_i = 0, \end{cases}$$

up to additive constants. At balance, its dynamics coincide with the $\alpha = 0$ reduced flow up to a factor of two in time. Away from balance, the mirror geometry depends on the conserved, gauge-dependent imbalance. The mobility $p_i^{1-\alpha}$ prescribed by Gauge-Muon depends only on the invariant path mass. Together with the capacity budget, this shows how α controls both the available path motion and the geometry of the reduced model, without implying a generalization advantage for any exponent.

F.6 Share and log-variance balance laws

Consider a continuous-time path flow with $p_i(t) > 0$, such as the reduced dynamics $\dot{p}_i = \nu_i p_i^{1-\alpha}$ of Lemma F.4, where ν_i is the demand the task places on path i. Define the total path mass $P = \sum_{j=1}^{h} p_j$, the path shares $\pi_i = p_i/P$, the log-masses $\ell_i = \log p_i$ with mean $\bar{\ell} = \frac{1}{h}\sum_i \ell_i$, and the log-growth rates $\psi_i = \dot{\ell}_i = \dot{p}_i/p_i$ (under these dynamics, $\psi_i = \nu_i p_i^{-\alpha}$).

Lemma F.7 (Share and log-variance balance laws)**.** *The path shares follow the replicator equation*

$$\dot{\pi}_i = \pi_i\Big(\psi_i - \sum_j \pi_j \psi_j\Big), \tag{55}$$

and the log-path variance $V_{\log p} = \frac{1}{h}\sum_i (\ell_i - \bar{\ell})^2$ *satisfies*

$$\frac{d}{dt} V_{\log p} = 2\mathrm{Cov}_i(\log p_i, \psi_i), \tag{56}$$

where $\mathrm{Cov}_i(a_i, b_i) = \frac{1}{h}\sum_i (a_i - \bar{a})(b_i - \bar{b})$ *is the uniform covariance over hidden units.*

Proof. Since $\dot{p}_i = p_i\psi_i$,

$$\dot{P} = \sum_i \dot{p}_i = \sum_i p_i\psi_i = P\sum_i \pi_i\psi_i. \tag{57}$$

Differentiate $\pi_i = p_i/P$:

$$\dot{\pi}_i = \frac{\dot{p}_i P - p_i \dot{P}}{P^2} \tag{58}$$

$$= \frac{p_i\psi_i}{P} - \frac{p_i}{P}\sum_j \pi_j\psi_j \tag{59}$$

$$= \pi_i\left(\psi_i - \sum_j \pi_j\psi_j\right), \tag{60}$$

which proves Equation 55.

Next, $\dot{\ell}_i = \psi_i$ and $\dot{\bar{\ell}} = \bar{\psi} := \frac{1}{h}\sum_i \psi_i$. Differentiating the variance gives

$$\frac{d}{dt} V_{\log p} = \frac{2}{h}\sum_i (\ell_i - \bar{\ell})(\psi_i - \bar{\psi}) = 2\mathrm{Cov}_i(\log p_i, \psi_i), \tag{61}$$

which proves Equation 56. Note that $\sum_i (\ell_i - \bar{\ell}) = 0$, so the $\bar{\psi}$ term contributes nothing and the derivative also equals $\frac{2}{h}\sum_i (\ell_i - \bar{\ell})\psi_i$. □

G Experimental Results

Compute. All experiments were run on a single NVIDIA RTX PRO 6000 GPU.

G.1 Why Muon breaks symmetry in ReLU networks

We would like to note here that the problem is not that Muon optimizer is "badly scaled". It is that two sets of parameters (U, V) and (DV, VD^{-1}) encoding the exact *same* function can produce *different* polar matrices. When a parameter coordinate is enlarged, the corresponding gradient coordinate shrinks. A symmetry-compatible optimizer must account for this contravariant transformation before applying its matrix oracle. To solve this, optimimizers must satisfy an exact computing diagram that rescale-then-optimize must equal optimize-then-rescale. Path-SGD (Neyshabur et al., 2015a) directly targets ReLU rescaling invariance. PathCond (Lebeurrier et al., 2026) chooses a favorable orbit representative. In contrast, our goal in this paper is different. That is, we want to retain Muon's spectral oracle while making the covector (U, V) exact invariant. LoRA-Muon (Cesista et al., 2026) addresses a related factor-rescaling issue for low-rank adaptation, but for a different matrix-factorization symmetry.

Example 1 Let $\phi(u, v) = \frac{1}{2}(uv - y)^2$ with scalar parameter u and v. At a minimizer, the only nonzero Hessian eigenvalue is $u^2 + v^2$. Suppose we transform $(u, v) \mapsto (cu, v/c)$ which preserves the output of $\phi(u, v)$, this eigenvalue becomes $c^2u^2 + c^{-2}v^2$. It will diverges as $c \to 0$ or $c \to \infty$ and is minimized at $c^2 = v/u$. Therefore, there is Euclidean learning rate that is uniformly stable across all the representation of on function.

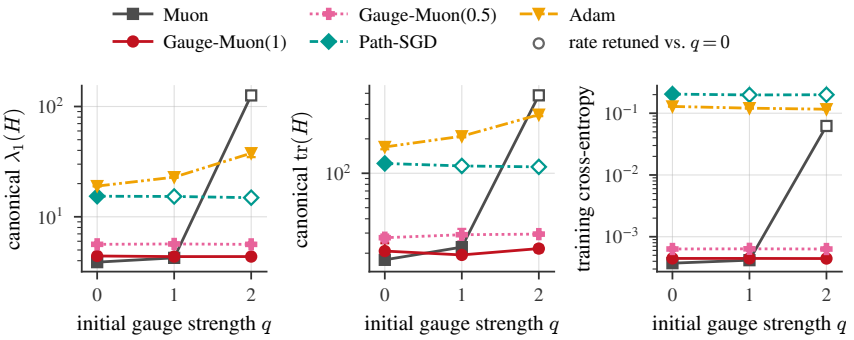


Figure 7: Canonical-gauge top Hessian eigenvalue (left), trace (middle) and training loss (right) at the solution each optimizer reaches, as a function of the gauge strength q at initialization. Sharpness is measured after canonicalizing each checkpoint to the balanced gauge (making $r_i = s_i$ for every i).

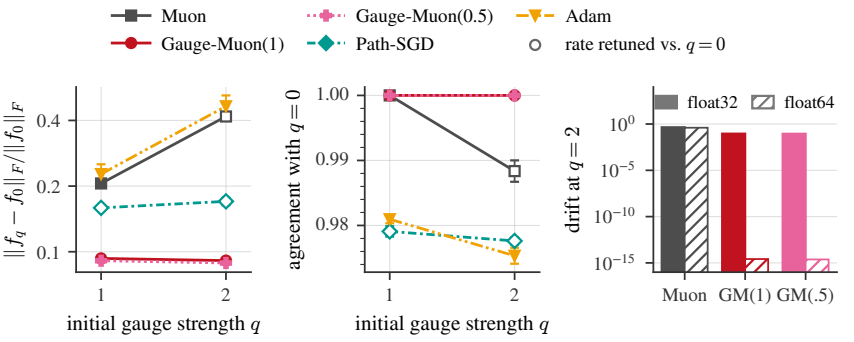


Figure 8: Change of the learned function when only the initial gauge changes. Left: relative drift from the $q = 0$ run at matched seed. Middle: agreement of predictions with the $q = 0$ run. Right: drift at $q = 2$ in float32 and float64 after 30 epochs.

Loss landscape between optimizers. We trained a two-layer ReLU network on FashionMNIST with optimizers including Muon, Adam, Path-SGD and Gauge-Muon(α), $\alpha \in \{0.5, 1\}$. We setup the initial gauge $d_i = 10^{qz_i}$, $q \in \{0, 1, 2\}$ that constructs the diagonal matrix D. Since the loss depends on (U, V) only through the predictor, all optimizers descend one and the same surface.

Sharpness. We consider to measure the sharpness of loss landscape by $\lambda_1(H)$, the largest eigenvalue of the Hessian $H = \nabla^2_{(U,V)}\mathcal{L}$ of the training loss with respect to all parameters (Dinh et al., 2017). Figure 7 shows the sharpness of all optimizers under different gauge initialization. We observe that Gauge-Muon is the only optimizer that reaches a low-loss *and* flat solution. At $q = 0$, Muon and Gauge-Muon(1) reach comparable solutions. The difference appears once the gauge is perturbed. At $q = 2$, Muon needs a smaller rate, stalls at a much higher loss and ends in a basin about $30\times$ sharper. Adam's sharpness also grows with q. Path-SGD is gauge-equivariant, so its sharpness does not change with q, however it needs its rate retuned and stalls at a high loss in a basin about three times sharper than Gauge-Muon's. Gauge-Muon(1) keeps the same rate, loss and sharpness at all three q. This suggests equivariance alone therefore does not produce a good minimum, and Muon's spectral approach alone does not survive a change of gauge. Gauge-Muon is designed to have both properties.

The same holds for the learned function (Figure 8). At matched seed, we measure the relative drift $\|f_q - f_0\|_F / \|f_0\|_F$ on a held-out probe batch. For Muon and Adam it grows with q and is two to five times larger than for Gauge-Muon, i.e. a large part of their solution is an artifact of the initial parametrization. Gauge-Muon's drift is small and does not grow with q.

G.2 Theoretical validation

We test four predictions of the theory on the bias-free two-layer ReLU network of width $h = 512$ on FashionMNIST. Unless stated otherwise, runs use three seeds, and the initial gauge is $d_i = 10^{qz_i}$ as in Appendix G.1.

Equivariance protects the learning rate. Theorem 4.2 states that Gauge-Muon's trajectory in function space does not depend on the representative. To see whether this matters in practice, we apply a function-preserving rescaling at the midpoint of training, either a LoRA-style merge that folds a factor of 10 into one side of every hidden unit, or a per-channel scale folding $d_i = 10^{2z_i}$ as done before post-training quantization, and sweep the learning rate over 10^{-4} to 0.316. Figure 9 shows that Gauge-Muon's entire learning-rate curve is reproduced under both perturbations. For Muon, the same perturbations move the optimal rate, raise the best loss, and make large rates far more damaging: at $\eta = 0.1$ the test loss rises from 0.51 to 1.87 and 5.03. Hidden-unit imbalance itself is not the problem: every optimizer, including SGD, drifts far from the balanced representative during training, and Gauge-Muon drifts the most. What separates the methods is whether the update depends on that position, which is a gauge coordinate rather than a property of the function.

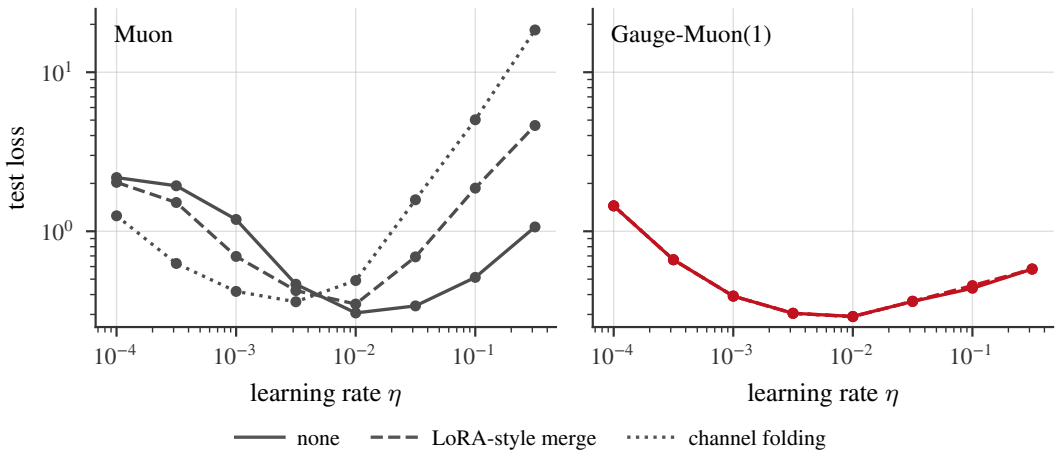


Figure 9: Test loss against learning rate after a function-preserving rescaling at the midpoint of training (FashionMNIST, $h = 512$, mean over three seeds). For Gauge-Muon(1) the three curves coincide.

Path mass is invariant and obeys the capacity budget. The total path mass $P(\theta) = \sum_i p_i$ is gauge invariant and controls generalization (Theorem F.1), so under Gauge-Muon its trajectory should not depend on q. We train for eight epochs at $q \in \{0, \dots, 4\}$ with fixed learning rates and measure the largest relative deviation of $P(\theta_t)$ from the $q = 0$ run. Figure 10 shows that Gauge-Muon holds $P(\theta_t)$ to float32 drift across eight orders of magnitude of rescaling (relative deviation at most $2times10^{-3}$), whereas Muon and AdamW move it by factors of up to 484 and 355. The growth of path mass tracks the loss: between $q = 0$ and $q = 4$ the final path mass of Muon grows $58\times$ and its test loss rises from .296 to 9.17, while Gauge-Muon(1) stays at a path mass of 472 and a test loss of .278. The capacity bound of Theorem F.3 holds on all 45 Gauge-Muon runs, with the bound between 6.2 and 7.0 times the measured growth of path mass, so it is valid but conservative at this scale.

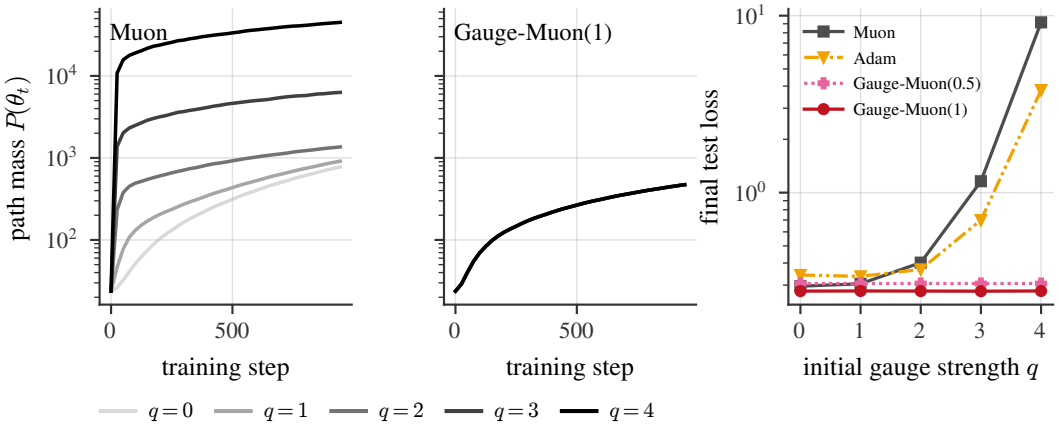


Figure 10: Path mass and test loss under a function-preserving initial gauge $d_i = 10^{qz_i}$ (FashionMNIST, $h = 512$, mean over three seeds). Left and middle: path-mass trajectory $P(\theta_t)$ for each q; for Gauge-Muon(1) the five curves *coincide*. Right: final test loss against q.

Path demand follows a power law in path mass. Lemma F.4 writes the first-order growth of each path as $\eta\nu_i p_i^{1-\alpha}$ and bounds the demand by $|\nu_i| \leq 2$. We record ν_i every ten steps for $\alpha \in \{0.5, 1, 1.5\}$. Late in training, $\log p_i$ explains a large share of the variance of $\log \nu_i$ (R^2 between 0.44 and 0.63) but almost none of the variance of the balanced covector norm (R^2 between 0.006 and 0.19). The structure therefore lives in the demand that the geometry exposes, not in the raw gradient, which is what separates Gauge-Muon from Path-SGD. The bound holds with margin: the largest measured $|\nu_i|$ is 0.72, and no unit exceeds 2 at any sample.

The tuned learning rate depends on the trajectory. A step is controlled by the trust ratio $\rho_i = \eta p_i^{-\alpha}$, which Theorem F.3 requires to stay bounded. If the tuned learning rate keeps this ratio fixed at a typical path mass $\tilde{p}$, then $\eta^* \propto \tilde{p}^{\alpha}$. Evaluated at the initial $\tilde{p}$, this gives a slope $d \log \eta^* / d\alpha = -3.1$, a $100\times$ fall in η^* over $\alpha \in [0, 1.5]$. The measured slope is -0.5, a fall of only $2\times$. The difference comes from $\tilde{p}$, which grows by one to two orders of magnitude during training, and faster for larger α. Averaged over the trajectory, the predicted slope lies between -0.5 and -1.9, which brackets the measurement, so the prediction holds once $\tilde{p}$ is taken along the trajectory rather than at initialization.

G.3 Numerical robustness

We test the numerical robustness of Gauge-Muon with respect to the accuracy of the Newton-Schulz polar approximation and reduced-precision computation. Table 1 reports the results at $\alpha = 1$ and $\eta = .003$. Three Newton-Schulz iterations can provide a good approximation, but increasing the number of iterations from three to seven improves the loss from .2960 to .2778 and substantially increases effective path utilization, with N_{eff}/h rising from .880 to .981. Seven iterations closely match the exact SVD baseline, which achieves a loss of .2765 and $N_{\text{eff}}/h = .982$. Therefore, a more

accurate polar approximation primarily improves optimization quality and path utilization, while seven Newton-Schulz iterations are sufficient to recover nearly all of the benefit of the exact SVD.

Reduced-precision computation is similarly robust. Using `bf16` autocast gives a loss of .2775, close to both the seven-step Newton-Schulz and exact-SVD variants, although its effective path utilization is moderately lower at $N_{\text{eff}}/h = .948$. Across all numerical variants, the loss gap between the gauge-equivalent $q = 0$ and $q = 4$ parameterizations remains small, ranging from 2.0×10^{-4} to 1.0×10^{-3}. This indicates that the practical numerical approximations preserve the gauge robustness of Gauge-Muon even when the resulting parameter trajectories are not numerically identical.

Table 1: Numerical robustness of Gauge-Muon at $\alpha = 1$ and $\eta = .003$. Probe drift denotes the relative prediction deviation between paired $q = 0$ and $q = 4$ runs.

variant	$q = 0$ loss	loss gap	probe drift	N_{eff}/h
Newton–Schulz (3 steps)	.2960	2.0×10^{-4}	0.025	0.880
Newton–Schulz (7 steps)	.2778	1.0×10^{-3}	0.051	0.981
exact SVD	.2765	5.9×10^{-4}	0.049	0.982
`bf16` autocast	.2775	7.0×10^{-4}	0.044	0.948

G.4 Image Classification Task

Table 2 reports the learning-rate grids that we used for model selection.

Table 2: Learning-rate search grids for the image-classification experiments, given as span (number of grid points) on the $\{1, 3\} \times 10^{-k}$ ladder. Muon and both Gauge-Muon variants share one grid per cell. The fine-tuning setting shifts every grid down by two steps.

Seting	Muon, Gauge-Muon	AdamW	Path-SGD
FashionMNIST MLP	0.0003–0.03 (5)	0.0001–.03 (6)	.1–30 (6)
CIFAR-100 ResNet-18	0.0003–0.03 (5)	0.0001–.01 (5)	0.03–10 (6)
CIFAR-100 ResNet-50	0.0003–.03 (5)	0.0001–.01 (5)	.01–10 (7)
SVHN ResNet-18	0.0003–.03 (5)	0.0001–.01 (5)	0.03–10 (6)
SVHN ResNet-50	0.0003–.03 (5)	0.00003–.01 (6)	0.003–10 (8)
CIFAR-100 ViT-S/16 FT	0.00003–0.003 (5)	0.00001–.001 (5)	0.00001–.03 (8)
CIFAR-100 ViT-B/16 FT	0.00003–0.003 (5)	0.00001–.001 (5)	0.00001–.03 (8)

G.4.1 Architectures

We provide the architecture used in our image classification task. We use 2-layer ReLU classifier

$$f(x) = V\,\mathrm{ReLU}(Ux), \qquad U \in \mathbb{R}^{h \times d_{\text{in}}}, \quad V \in \mathbb{R}^{C \times h}, \tag{62}$$

Both matrices use Kaiming-uniform initialization with no bias. We use the hidden width as $h = 512$. For width ablation experiments, we sweeps $h \in \{128, 256, 512, 1024, 2048\}$.

MLP (FashionMNIST) The classifier is applied directly to raw pixels, so $d_{\text{in}} = 784$ and $C = 10$.

ResNet-18 and ResNet-50 (CIFAR-100, SVHN) The colour datasets enter through a *frozen* ImageNet-pretrained ResNet. We trained only the two-layer head and freeze the backbone.

ViT-S/16 and ViT-B/16 (CIFAR-100) We perform full fine-tuning ViTfrom the SpikeZIP-TF ReLU checkpoints[2] (You et al., 2024). In this version, GELU activations were replaced by ReLU before further ImageNet-1k training (82.3% top-1 for ViT-S/16, 83.5% for ViT-B/16). The pretrained classifier is removed and randomly initialized. For data augmentation, we augmented CIFAR-100

[2] https://huggingface.co/XianYiyk/SpikeZIP-TF-vit-base-patch16-relu and https://huggingface.co/XianYiyk/SpikeZIP-TF-vit-small-patch16-relu

images with a random horizontal flip and a reflect-padded random crop, then upsampled bilinearly to 224×224 following the transfer protocol of Dosovitskiy et al. (2020).

Table 3 collects all the detailed information of architectures for the image classification task.

Table 3: Architecture of each cell. d_{in} is the input dimension of the trained two-layer ReLU pair; for the fine-tuned transformers it is the MLP hidden width of each of the twelve blocks. Split sizes are train/validation/test.

dataset	backbone	trained part	d_{in}	split
FashionMNIST	none (raw pixels)	two-layer ReLU	784	54,000 / 6,000 / 10,000
CIFAR-100	frozen ResNet-18	two-layer ReLU	512	45,000 / 5,000 / 10,000
CIFAR-100	frozen ResNet-50	two-layer ReLU	2048	45,000 / 5,000 / 10,000
SVHN	frozen ResNet-18	two-layer ReLU	512	65,931 / 7,326 / 26,032
SVHN	frozen ResNet-50	two-layer ReLU	2048	65,931 / 7,326 / 26,032
CIFAR-100	ReLU ViT-S/16	all, MLP pairs gauged	1536	45,000 / 5,000 / 10,000
CIFAR-100	ReLU ViT-B/16	all, MLP pairs gauged	3072	45,000 / 5,000 / 10,000

G.4.2 Learning-rate sensitivity

Figure 11 plots tuning validation accuracy against the learning rate divided by the selected learning of each method. Here, all curves peak at 1 on the horizontal axis to make their *shapes* directly comparable.

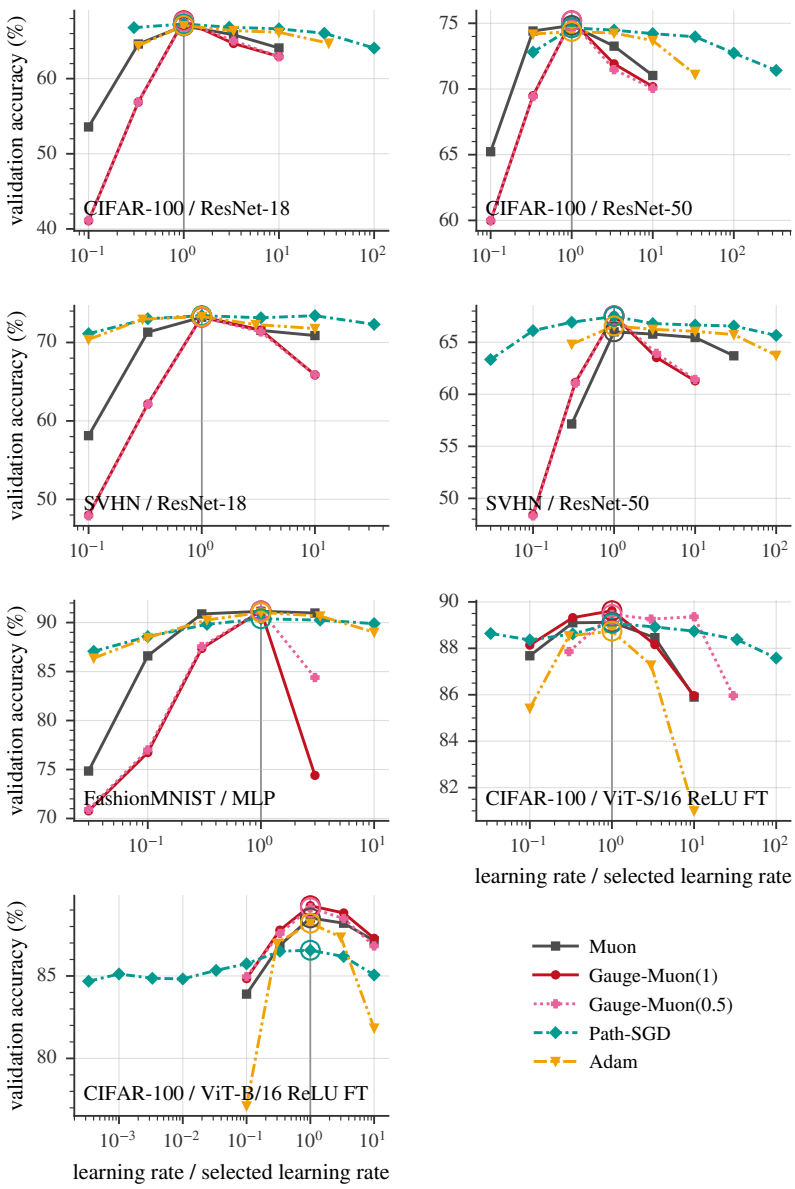


Figure 11: Learning-rate sensitivity for all seven settings. Tuning validation accuracy is plotted against the learning rate divided by each method's own selected rate, so every curve peaks at 1 on the horizontal axis and the shapes are directly comparable; rings mark the selected rates.

G.4.3 Training curves

Figure 12 shows the validation accuracy per epoch at the selected learning rate of each method. Figure 13 provides the corresponding training loss on a log axis. The bands in each plot are the standard errors over seeds.

Discussion We observe that Gauge-Muon tends to fits later in the accuracy curves and takes over Muon near the end. Other hand, the loss curves reveals that Gauge-Muon may have higher training loss but still maintain higher accuracies. This may suggest that Gauge-Muon reaches a comparable or better solution while fitting the training set markedly less. This behavior is expected from a method that constrains path mass rather than merely descending faster.

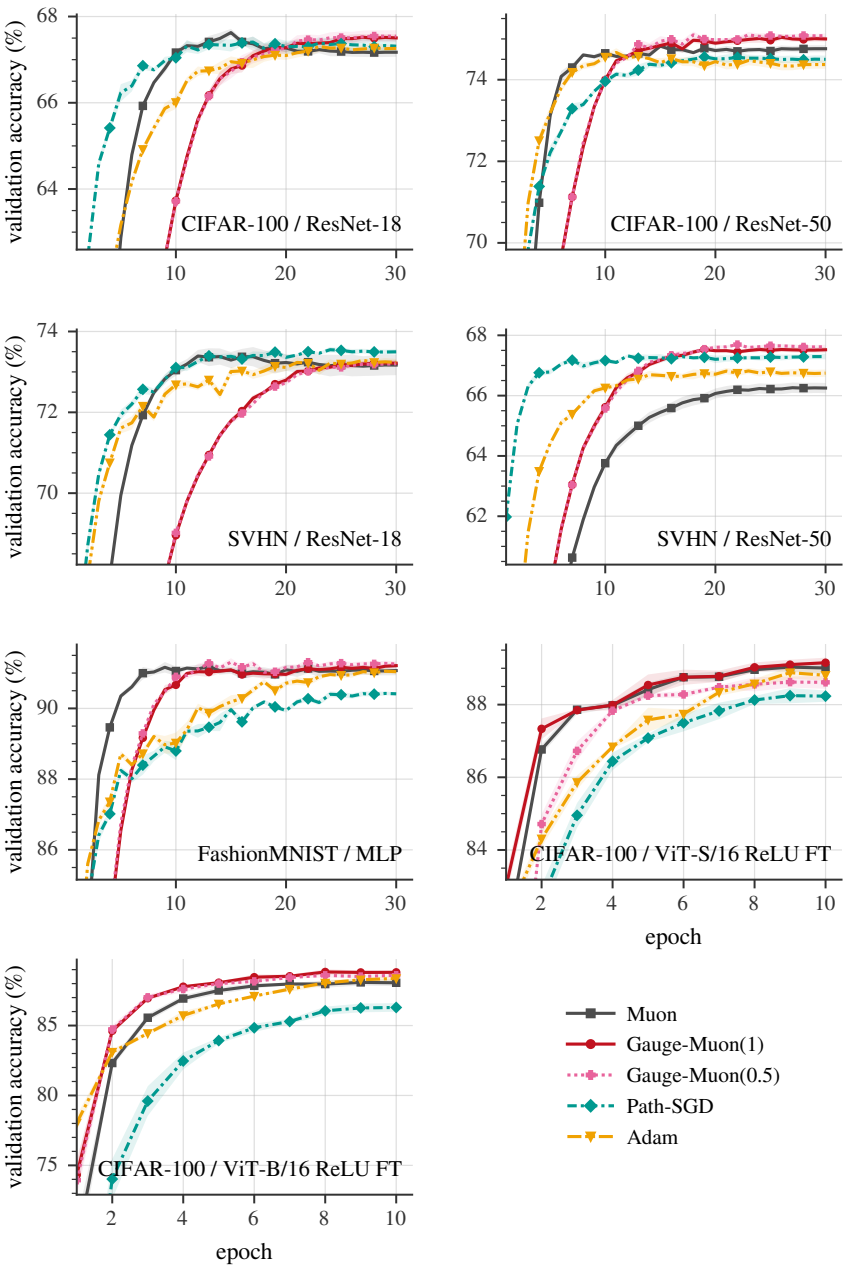


Figure 12: The validation accuracy per epoch at each method's selected learning rate, with standard-error bands over seeds. Each panel is scaled to its converged region, so the first epochs run off the bottom of the axes.

Comparing to PathSGD and Adam On FashionMNIST, spectral methods (Gauge-Muon and Muon) make training loss to about 4×10^{-4} while Path-SGD and Adam stall at 0.210 and 0.132. On the ViT setting, Path-SGD is underperformed, ending at a training loss of $.231$ against 8.4×10^{-3} for Gauge-Muon at $\alpha = 1$. This is a clear evidence that our Gauge-Muon wins over PathSGD in this benchmark.

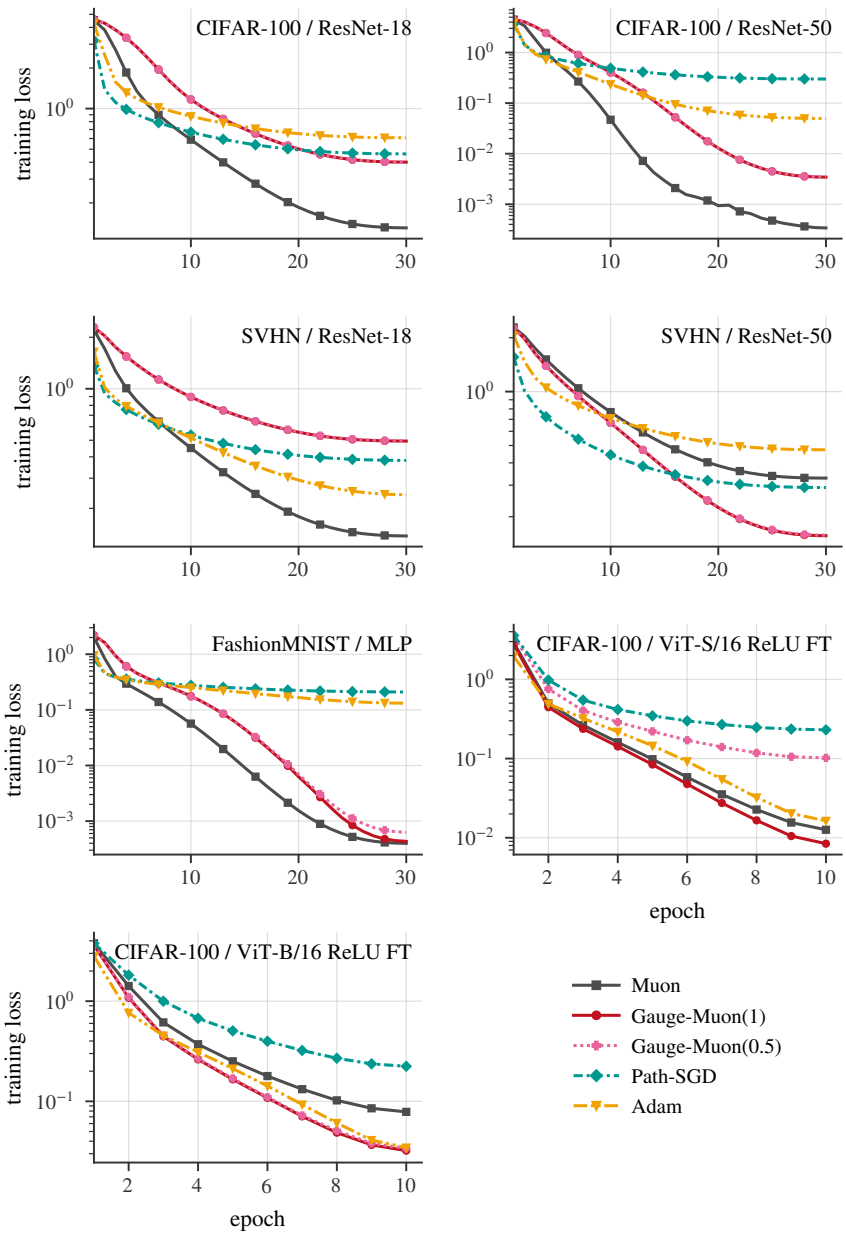


Figure 13: Mean training loss per epoch (log scale) at each method's selected learning rate, with standard-error bands over seeds.

G.5 Language Modeling Task

G.5.1 Architectures

We used the same GPT-2 small setup for our experiments in the language modeling tasks including $d_{\text{model}} = 768$, pre-norm, a 512-token context and GPT-2 tokenizer. We only change the feed-forward block which is the main focus of Gauge-Muon.

In our main text, we consider two settings of feed-forward blocks including dense feed-forward and mixture of experts.

Dense feed-forward Each block has incoming and outgoing matrices form the pair $(W_{\text{up}}, W_{\text{down}})$ that Gauge-Muon is defined on. There are two options for our activations: ReLU and SwiGLU.

For the ReLU variant,the homogeneous theory covers directly, and we set the mutiplicative factor between hidden unitand hidden dimension as $h/d = 4$ ($h = 3072$). For the SwiGLU variant, we used $h/d = 8/3$, so $h = 2048$ because it has an extra gate matrix W_{gate} alongside W_{up}.

Mixture of experts The mixture of experts (MoE) settings replace each dense MLP with eight experts under top-1 routing. The ReLU expert width is 384 while the SwiGLU expert width is 256. This ensures the MoE settings have the same parameters as the dense settings. The router is a plain matrix trained with Adam.

For load balancing, we use a Switch-style load-balancing auxiliary (Fedus et al., 2022) to add to the optimization objective at weight 0.05. This is because at weight 0 and at 10^{-3} routing collapses onto a single expert.

Table 4: Feed-forward architecture of the language-model settings. All setting share the GPT-2 small backbone (12 layers, $d_{\text{model}} = 768$, 12 heads, context 512, tied embeddings). h is the hidden width of one MLP, or of one expert in the mixture settings.

Setting	activation	h	gauged pairs	note
dense	ReLU	3072	12	$h/d = 4$
dense	SwiGLU	2048	12	$h/d = 8/3$, gate shares the hidden index
mixture	ReLU	384	96	8 experts, top-1
mixture	SwiGLU	256	96	8 experts, top-1

G.5.2 Training setup and optimizer

All runs train on FineWeb-Edu, tokenised with the GPT-2 BPE, at a batch of 64 sequences with 4 gradient-accumulation steps and a context of 512, giving $131{,}072$ tokens per step. In our experiments, we consider two token budget settings: low-budget (150M tokens) and high-budget (1B tokens). The corresponding trainig steps are $1{,}100$ and $7{,}600$ steps. The optimizer scheduler is cosine with 500 warmup steps, gradient clipping at 0.5.

We split parameters by their role, as summarized in Table 5. As we focus on studying the gauge symmetry in feed-forward blocks. Everything else is set with the same settings.

Table 5: Assignment of parameter groups to optimizers. Only the first row changes between arms; the other two are identical in every run, so the swept rate η_{MLP} governs the feed-forward blocks alone.

parameter group	optimizer	learning rate	settings
MLP pairs (expert pairs in MoE)	the swept arm	η_{MLP}, tuned	see Table 6
attention matrices	Muon	10^{-2}, fixed	momentum .95, Newton–Schulz 5
embeddings, LayerNorm, router	AdamW	10^{-3}, fixed	$\beta = (.8, .999)$, weight decay=.1

Table 6 lists the six settings of our experiments for language model. We try our best to tune learning rates for each setting rather than sharing a default learning rate. Here, we note that Path-SGD's ratio match rescales its update to the raw gradient norm.

Table 6: The six arms, their settings on the MLP pairs, and the range of rates they select across the settings in this section. The selected rates span four decades, which is why a shared grid would make the comparison meaningless.

arm	settings on the MLP pairs	selected η_{MLP}
Muon	momentum .95, Newton–Schulz 5	.0025–.02
Gauge-Muon $\alpha = 1$	geometric-mean gauge	.005–.02
Gauge-Muon $\alpha = .5$	geometric-mean gauge	.005
Path-SGD	momentum .9, ratio match	.64–40.96
Adam	$\beta = (.8, .95)$, no weight decay	.001–.004

G.5.3 More Results

Figure 14 and Table 7 summarize all 32 settings. At 150M tokens Gauge-Muon beats Muon in every setting (-1.8% to -4.0%). At 1B tokens the result splits by architecture: Gauge-Muon stays ahead in both mixtures (-2.46%, -2.33%) but falls behind in both dense models ($+1.01\%$, $+1.90\%$). Activation changes the size of the delta, not its sign. Path-SGD, although gauge-equivariant, trails Muon everywhere, so equivariance alone does not produce the gain.

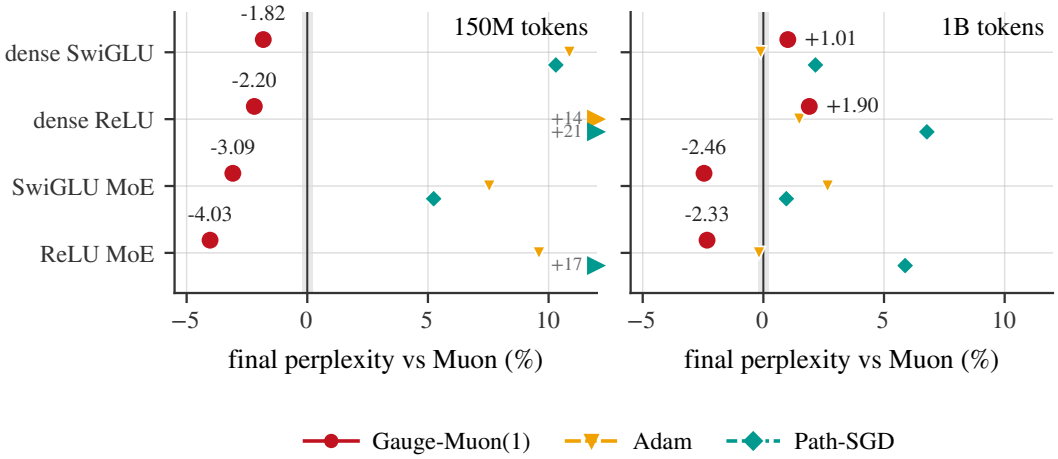


Figure 14: Final validation perplexity relative to Muon, every optimizer at its own tuned η_{MLP}. Negative favours the optimizer; values beyond $+12\%$ are clipped and labelled.

Table 7: Final validation perplexity at each optimizer's tuned rate (in parentheses η_{MLP}). Best of the listed columns in bold; Gauge-Muon(0.5), run only on ReLU MoE at 1B, reaches 27.93 at .005. Perplexities are comparable only within a row.

budget	setting	Muon	Gauge-Muon(1)	Adam	Path-SGD
150M	dense SwiGLU	41.47 (.015)	**40.72** (.02)	45.98 (.001)	45.75 (2.56)
	dense ReLU	43.71 (.01)	**42.75** (.02)	49.94 (.002)	52.79 (5.12)
	SwiGLU MoE	44.81 (.01)	**43.42** (.01)	48.18 (.001)	47.15 (.64)
	ReLU MoE	45.35 (.01)	**43.52** (.02)	49.70 (.001)	52.92 (.64)
1B	dense SwiGLU	25.89 (.02)	26.15 (.01)	**25.86** (.004)	26.45 (40.96)
	dense ReLU	**26.55** (.02)	27.05 (.01)	26.94 (.002)	28.34 (10.24)
	SwiGLU MoE	28.08 (.0025)	**27.39** (.005)	28.82 (.001)	28.34 (1.28)
	ReLU MoE	28.64 (.01)	**27.97** (.005)	28.59 (.002)	30.32 (10.24)

Figure 15 shows the full learning-rate sweeps behind these numbers. Gauge-Muon's optimum is sharp, with perplexity rising steeply one grid step above it, whereas Muon's is flat over a wider range of rates; Appendix G.5.5 traces this to the compounding of path mass under Gauge-Muon's relative step. Path-SGD's tuned rate varies by almost two orders of magnitude across settings, which is why every optimizer is tuned on its own grid.

G.5.4 Ablations

Ingredients Figure 16 (left) changes one ingredient at a time in the SwiGLU MoE at 150M and measures each variant against Muon in the same setting. The gain does not depend on the exact exponent: smaller values of α keep it. Two changes remove it. Dropping the geometric-mean normalization turns the gain into a loss, even after retuning the learning rate, and does so in the dense $h = 256$ model as well, which has no router; the normalization is therefore necessary in both settings. Training the router with SGD instead of AdamW makes the two optimizers tie. Muon is barely affected, while Gauge-Muon loses its whole gain.

Seeds Across three seeds (Figure 16, right), the worst Gauge-Muon seed (44.49) still beats the best Muon seed (44.81); the mean gain is 2.92%.

Comparability Muon's rectangular scale factor $\max(1, \text{rows}/\text{cols})^{1/2}$ equals 1 for an expert's 384×768 incoming matrix but 2 for the dense 3072×768. Every optimizer within the mixture shares that factor, so comparisons *inside* a panel are unaffected, but mixture perplexities are not directly comparable to dense ones.

G.5.5 Why narrow blocks and short budgets favor Gauge-Muon

In this section, we expand our inverstiation in Figure 6.

Relative steps make path mass compound Gauge-Muon(1) moves every hidden unit by a fixed fraction of its own norm, so by Lemma F.4 path mass grows exponentially in the integrated learning rate; Muon's step has a fixed absolute size, so its path mass grows polynomially (Figure 6(a)). Every tuned Gauge-Muon setting sits one grid step below the rate at which growth explodes and perplexity degrades, whereas Muon's optimum is flat over a four- to eight-fold range. Since the admissible growth caps ηT, Gauge-Muon's tuned rate halves from 150M to 1B tokens in the settings tuned at both budgets, and at 1B it is at most half of Muon's in every dense setting (Table 8). The smaller rate delays the crossing from 40–130M tokens at 150M to 500–900M at 1B, and to 1.77B for wide experts trained to 2B.

The aspect ratio sets Muon's allocation Let G be the $h \times d$ gradient of a block's incoming matrix. For $h \le d$ the polar factor of G has orthonormal rows, so Muon moves every unit by the same absolute amount. For $h > d$ its squared row norms are the leverage scores of G, which track the gradient row norms (correlation 0.9–1.0), so Muon moves units with larger gradients more. Gauge-Muon(1) gives every unit the same relative motion at every width. Hence Muon's spread of log path mass is larger in wide blocks than in narrow ones, while Gauge-Muon's stays small everywhere (Figure 6(b), Table 8). Our reading is that uniform relative motion is the right prior when every unit must carry part of the function, and that gradient-weighted allocation builds a working sub-network faster when the block is wide; the geometry follows from the polar factor, but no result in the paper says which geometry trains faster.

Table 8: Tuned rates, concentration and gap trajectory in the 1B-token settings. Spread is $\mathrm{Var}_i \log p_i$ at the end of training; the crossing is the first point after which the gap stays negative.

		tuned η		Muon		
Settings	h/d	Muon	Gauge-Muon	spread	crossing	final gap
dense $h = 256$	0.33	.01	.005	.016	577M	-1.2%
dense $h = 512$	0.67	.02	.005	.015	891M	-0.1%
SwiGLU MoE 8×256	0.33	.0025	.005	.008	498M	-2.5%
ReLU MoE 8×384	0.50	.01	.005	.021	603M	-2.3%
dense $h = 2048$	2.67	.02	.01	.032	never	$+1.0\%$
ReLU $h = 3072$	4.0	.02	.01	.036	never	$+1.9\%$

The two effects are separable The ramp alone does not explain the wide-block deficit: at 1B tokens Gauge-Muon's path mass reaches one at a similar time in narrow and wide blocks (262–511M tokens), yet only the narrow blocks cross. Conversely, the width effect exists at equal rates: at 150M tokens with both optimizers at $\eta = .01$, the gap is -2.0%, $+1.1\%$ and $+4.9\%$ at $h = 256$, 512 and 2048. Tuned rates hide this at 150M because the short budget lets Gauge-Muon run at a rate at least as large as Muon's.

The differences arise early Muon's absolute steps settle each unit's ratio of incoming to outgoing norm within the first 10–25% of training. Consistently, resuming a Muon checkpoint at 100M tokens for 50M more gives the same perplexity with either optimizer at $h = 256$, 768 and 2048 (within 0.15%). The gaps in Figure 6(c) are made while the network grows out of its initialization.

Thirty-two experts The expert series shares $\eta = .01$. At 32 experts of width 64, Gauge-Muon's path mass grows $240\times$ against $7.9\times$ at eight experts, above the ceiling where a dense block loses about 5% to its own optimum. We expect $\eta = .005$ or .0025 to recover the 16-expert gain; this has not been run.

Insensitivity to the exponent With the geometric-mean normalization, α acts only through the per-unit weights $(\tilde{p}/p_i)^\alpha$, whose spread is about $\exp(\alpha\sigma)$ for σ the standard deviation of $\log p_i$ within a layer. The runs keep $\sigma \le 0.18$, so for $\alpha \in [0.25, 1]$ the weights stay within about 15% of one, which is why $\alpha = 0.25$, 0.5 and 1 land within 0.5% of each other in Figure 16 and Gauge-Muon(0.5) ties Gauge-Muon(1) on the ReLU mixture.

G.6 Gauge-fixed quantization-aware training

Why a representative is required. Let ϕ be positively homogeneous of degree ν, $\phi(cz) = c^\nu \phi(z)$, and consider the feed-forward pair

$$f_{U,V}(x) = V\phi(Ux), \qquad (U, V) \sim (DU, VD^{-\nu}), \tag{63}$$

where $D = \mathrm{diag}(d_1, \dots, d_h) \succ 0$. Although the two parameterizations in Equation 63 compute the same function, ordinary weight quantization acts on their ambient coordinates and generally gives different functions. For example, per-output-channel quantization commutes with the row scaling of U, but not with the column scaling of V. A groupwise quantizer has the same asymmetry because its groups lie along each matrix's input dimension.

We use symmetric uniform fake quantization (Jacob et al., 2017). For a tensor or quantization group w,

$$Q_b(w) = \Delta\ \mathrm{clip}(\mathrm{round}(w/\Delta), -q_{\max}, q_{\max}), \qquad \Delta = \frac{\|w\|_\infty}{q_{\max}}, \quad q_{\max} = 2^{b-1} - 1. \tag{64}$$

The scale Δ is shared either by the full matrix, by one output row, or by a contiguous group of input coordinates (group size 128 by default).

Canonical representatives. Write $r_i = \|U_{i,:}\|$ and $s_i = \|V_{:,i}\|$, using either the ℓ_2 norm or the ℓ_∞ norm as indicated below. The implementation supports three data-free gauge choices:

$$d_i^{\mathrm{bal}} = (s_i/r_i)^{1/(1+\nu)} \qquad (\ell_2\text{-balanced}), \tag{65}$$

$$d_i^{\mathrm{eq}} = (s_i/r_i)^{1/(1+\nu)} \qquad (\ell_\infty\text{-equalized}), \tag{66}$$

$$d_i^{\mathrm{vcol}} = \frac{s_i^{1/\nu}}{\exp\left(h^{-1}\sum_j \log s_j^{1/\nu}\right)} \qquad (\text{equal } \ell_\infty \text{ column ranges in } V). \tag{67}$$

Equalization is best suited to a per-tensor grid, which must control the ranges of both matrices; `vcol` is suited to per-channel and groupwise grids, for which row scaling already commutes with quantization of U and only the columns of V remain exposed.

Let $D(U, V)$ denote one of these canonicalizing gauges and let $T_D(U, V) = (DU, VD^{-\nu})$. The pair quantizer is

$$\widetilde{Q}_b(U, V) = T_{D(U,V)^{-1}}\left(Q_b(DU), Q_b(VD^{-\nu})\right). \tag{68}$$

Because the canonical representative depends only on the gauge orbit (up to a harmless global scale), this construction is exactly covariant:

$$\widetilde{Q}_b(T_E(U, V)) = T_E \widetilde{Q}_b(U, V) \tag{69}$$

for every positive diagonal E, at tensor, channel, and group granularity. Thus the quantized predictor is a well-defined function on the quotient rather than on an arbitrary checkpoint representative. We map back in Equation 68 returns weights in the caller's gauge; omitting it instead returns the canonical weights expected at export. The two forms represent the same function and, under the straight-through estimator below, produce the same gradient.

QAT implementation. During quantization-aware training, each forward pass substitutes

$$W_{\mathrm{STE}} = W + \mathrm{stopgrad}(W_q - W), \tag{70}$$

so its value is quantized while its derivative with respect to the float32 master weight is the identity. Gauge scales and quantization scales are detached. Quantization is performed in float32 with autocast disabled, after which the surrounding autocast may use bf16 for the matrix multiplication. A non-STE quantizer is active only in evaluation mode; enabling hard rounding during training would make the weight gradient zero almost everywhere.

The same hook is attached to every $(W_{\mathrm{up}}, W_{\mathrm{down}})$ pair in a transformer MLP and independently to every expert in a mixture-of-experts layer. The homogeneity degree is taken from the model configuration ($\nu = 1$ for ReLU and SwiGLU, $\nu = 2$ for squared ReLU), since an incorrect degree silently breaks function preservation. For SwiGLU the continuous gauge acts on $(W_{\mathrm{up}}, W_{\mathrm{down}})$ only; W_{gate} is therefore quantized separately without gauge fixing. The optimizer continues to update the master parameters and is otherwise unaware of the quantizer.

G.6.1 Experiments

Setup. We use the bias-free two-layer ReLU network of width 512 on FashionMNIST, which has the shape of a transformer feed-forward block, and initialize it in the gauge $d_i = 10^{qz_i}$ with $q \in \{0, 1, 2\}$, as in Appendix G.1. Every optimizer is trained for 30 epochs with a cosine schedule; its learning rate is tuned on three seeds for each quantizer configuration and evaluated on three fresh seeds. We report W4 per-channel quantization in the main comparison; W8 and groupwise grids show the same pattern with smaller drops. Because moving a checkpoint along its gauge orbit is exactly function preserving, the float32 network is identical before and after gauge fixing.

Post-training quantization. Figure 17 (left) shows that the damage of rounding a checkpoint as trained grows with q, and grows fastest for the gauge-equivariant optimizers. This is the expected cost of equivariance: Gauge-Muon and Path-SGD keep the representative they were given, so an unbalanced initial gauge is still present at the end of training, whereas Muon and Adam partly rebalance it. Gauge fixing before rounding removes this dependence for every optimizer and costs nothing in float32 accuracy. With a quantizer that sees only a canonical representative, what remains is the quantization error of the function itself.

Quantization-aware training. Figure 17 (right) and Table 9 train through the quantizer. With the gauge-fixed quantizer, Gauge-Muon reaches the same accuracy at $q = 2$ as at $q = 0$ (within 0.2 percentage points), while Muon and Adam lose accuracy as the initial gauge is perturbed. The quantizer is now covariant, so the remaining sensitivity of Muon comes from the optimizer itself: combining a covariant quantizer with an equivariant optimizer makes the whole QAT pipeline independent of the initial gauge. Path-SGD is also unaffected by q, but trails the spectral methods at both values. Gauge fixing matters most for a per-tensor grid: without it, QAT at $q = 2$ fails for both equivariant optimizers, since a single scale must cover the widely spread rows and columns they preserve, while with equalization Gauge-Muon again matches its $q = 0$ accuracy.

Table 9: W4 quantization-aware training on FashionMNIST: test accuracy (%) for each quantizer configuration and initial gauge q, mean over three seeds. Standard errors are at most 0.23 except Gauge-Muon(0.5) with a per-tensor grid and no gauge at $q = 2$ (2.07). Best in each column in bold. Path-SGD's selected rate lies at the top of its grid.

	per-channel				per-tensor			
	no gauge		`vcol`		no gauge		equalized	
optimizer	$q=0$	$q=2$	$q=0$	$q=2$	$q=0$	$q=2$	$q=0$	$q=2$
Muon	90.05	88.73	90.19	89.33	90.01	**86.61**	90.06	88.93
Gauge-Muon(1)	**90.22**	**89.43**	**90.40**	90.23	**90.09**	10.00	90.11	90.08
Gauge-Muon(0.5)	90.13	89.20	90.18	**90.24**	89.74	70.95	**90.25**	**90.13**
Path-SGD	88.76	87.66	88.82	88.69	88.73	10.00	88.74	88.67
Adam	89.51	88.76	89.66	89.31	89.35	84.40	89.49	89.23

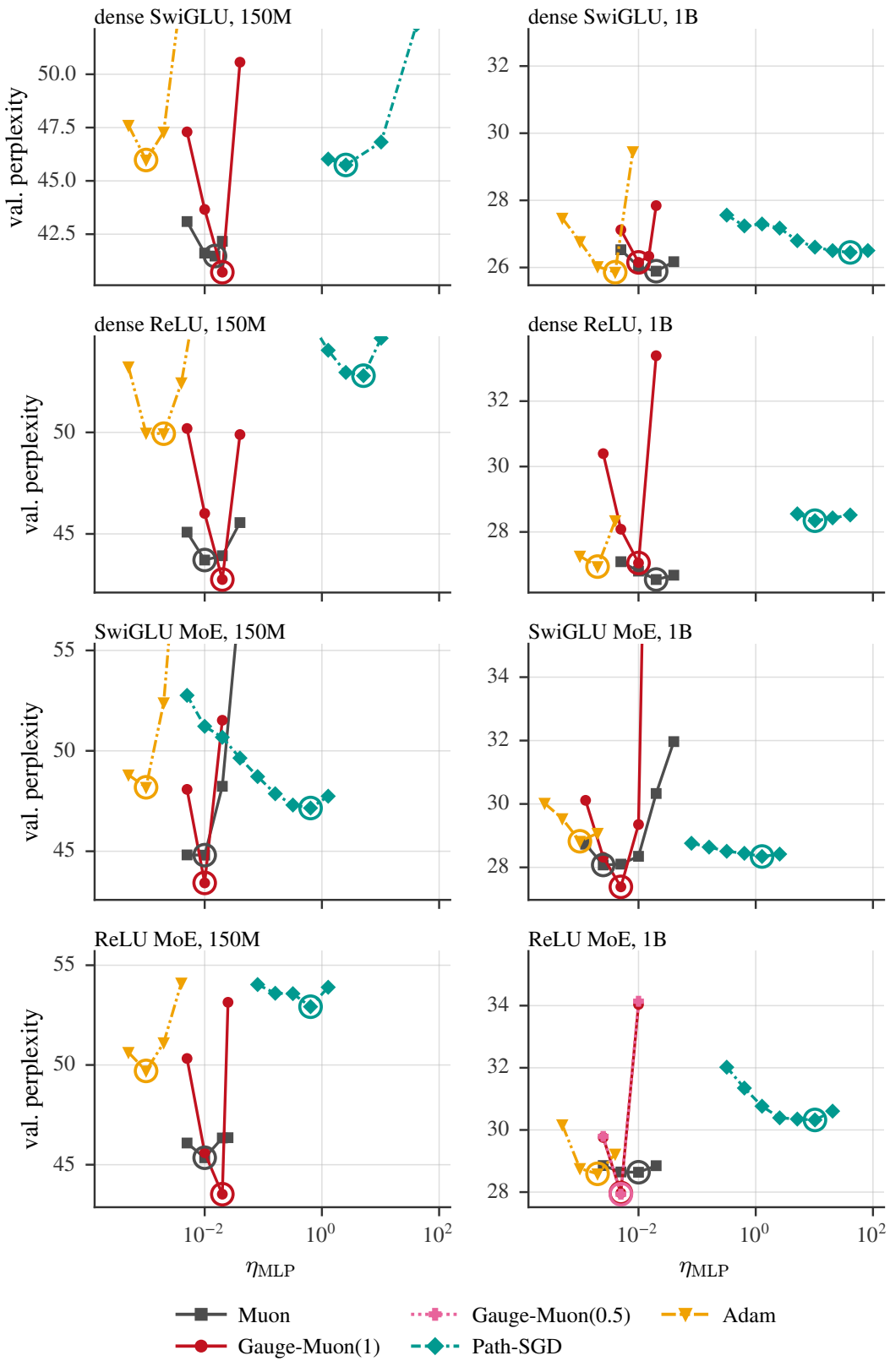


Figure 15: Validation perplexity against the MLP learning rate for every setting and budget. Rings mark each optimizer's tuned rate; a dotted ring marks an optimum at the edge of its grid. The vertical axis is cut at 28% above the best perplexity in each panel.

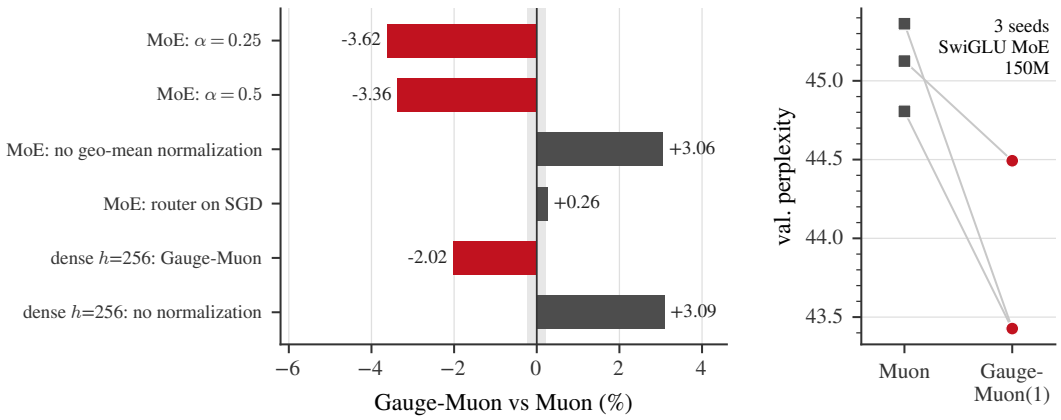


Figure 16: Left: Gauge-Muon's delta against Muon in the same setting after removing one ingredient (SwiGLU MoE, 150M; last two rows dense $h = 256$). Right: three seeds of the SwiGLU MoE at 150M, paired by seed.

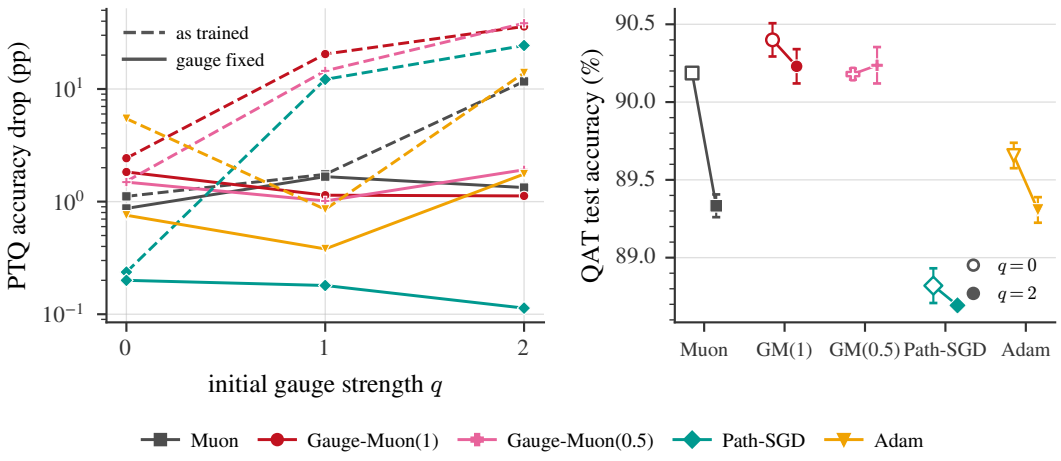


Figure 17: W4 per-channel quantization on FashionMNIST. Left: post-training accuracy drop against the initial gauge, rounding the checkpoint as trained (dashed) or after moving it to the `vcol` representative (solid). Right: accuracy after quantization-aware training with the gauge-fixed quantizer, at $q = 0$ (open) and $q = 2$ (filled); mean and standard error over three seeds.